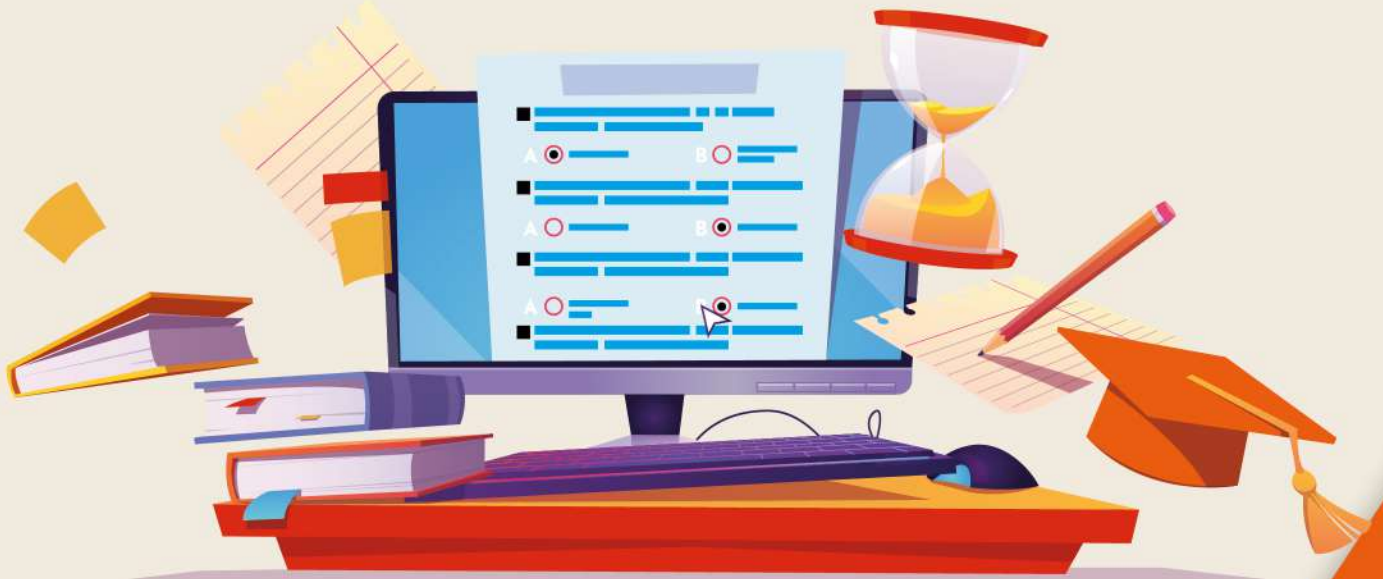


أسئلة سنوات فاينال

مختبر فيزياء عامة 1



بسم الله الرحمن الرحيم

نقدم لكم نحن أسرة فريق
(لجنة السنافر)

مجموعة أسئلة اختبارات إلكترونية تم تجميعها خلال الفصول الماضية
سائلين المولى أن يوفقنا وإياكم لكل خير

تنويه

يوجد بعض الأسئلة عليها إجابات قد تحتل الصواب وقد تحتل الخطأ
فإن أصبنا فما هو إلا توفيق من الله
وإن أخطأنا فمن أنفسنا

#خدمتكم_طريق_خضناه_لرضى_الله

#الإتجاه_الاسلامي

#بسواعدنا_نبنيها

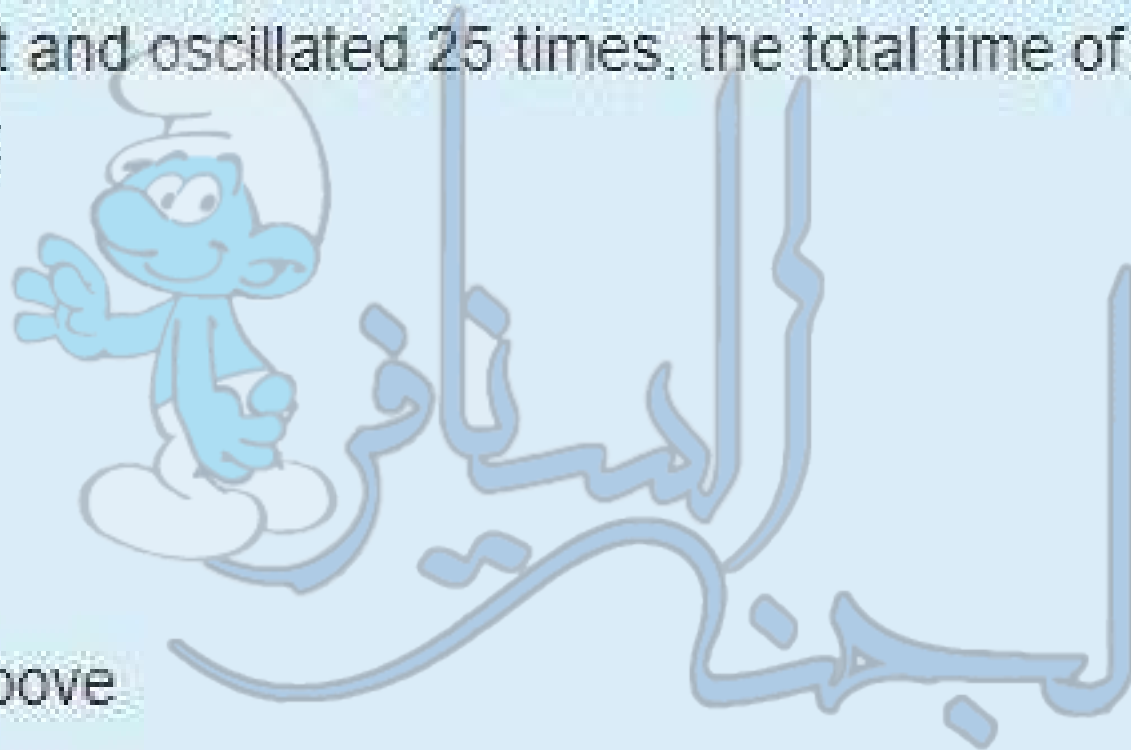
#لجنة_السنافر

#هي_الله

In the simple harmonic oscillator, a mass of (40g) is hanged in a spring, if the extension of the spring is (4 cm) and if the mass drawn to (3cm) from equilibrium point and oscillated 25 times, the total time of an oscillation of 20 times is (in sec):

Select one:

- ☒ a. 9.93
- ☐ b. 11.91
- ☐ c. Non of above
- ☐ d. 5.96
- ☐ e. 7.95 ✓



In a vector experiment, the ring being large or small does not affect the equilibrium



إختبر واحداً:

صح ☐

خطأ ☐

Two forces are in equilibrium, if you know that $(F_{1x} = 0.25 \text{ N})$, $(F_{1y} = -0.4 \text{ N})$ then the magnitude of (F_2) , and its deviation from the positive (X) axis

Select one:

- ☐ a. (0.47 N, 57.99°)
- ☐ b. (0.47 N, 237.99°)
- ☐ c. (0.47 N, 302.01°)
- ☐ d. (0.47 N, 122.01°)

Two forces are in equilibrium, if you know that ($F_{1x} = 0.25 \text{ N}$), ($F_{1y} = -0.4 \text{ N}$) then the magnitude of (F_2), and its deviation from the positive (X) axis



اختر أحد الخيارات:

a. (0.47 N, 57.99°)

b. (0.47 N, 237.99°)

c. (0.47 N, 302.01°)

d. (0.47 N, 122.01°)

In the simple harmonic oscillator, a mass of (40g) is hanged in a spring, if the extension of the spring is (4 cm) and if the mass drawn to (3cm) from equilibrium point and oscillated 25 times, the total time of an oscillation of 20 times is (in sec):

Select one:

- ☐ a. 9.93
- ☐ b. 11.91
- ☐ c. Non of above
- ☐ d. 5.96
- ☒ e. 7.95 ✓



The correct answer is: 7.95

In uniform circular motion, the speed is constant but the velocity is variable

Select one:

- ☒ a. True ✓
- ☐ b. False



The correct answer is: True

In spring constant experiment, if the hanging mass 80g and the elongation in the spring is 4cm, the spring constant (N/m) is:

Select one:

- ☐ a. 40
- ☒ b. 20 ✓
- ☐ c. 10
- ☐ d. Non of above
- ☐ e. 5

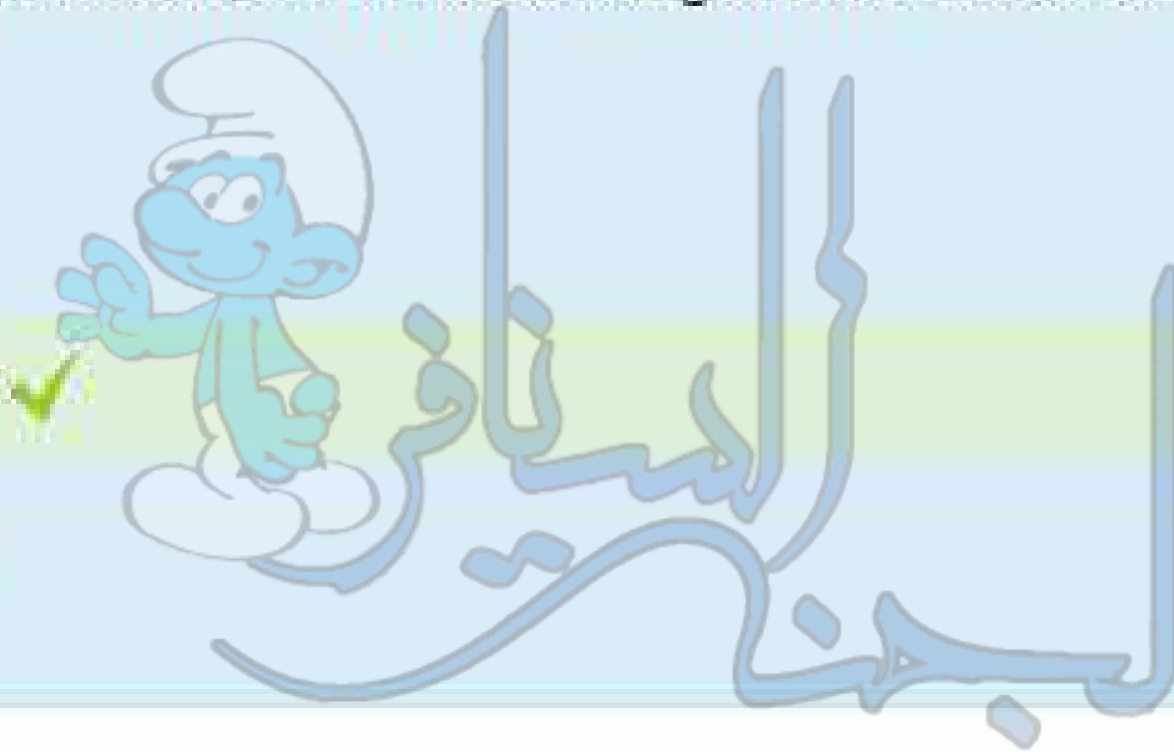


The correct answer is: 20

In vector experiment, the mass hangers do not affect on the equilibrium state

Select one:

- ☒ a. False ✓
- ☐ b. True



The correct answer is: False

We can conclude from the frictional force that μ_k is smaller than μ_s

Select one:

- ☒ a. True ✓
- ☐ b. False



The correct answer is: True

The micrometer is more accurate than the vernier caliper

Select one:

- ☐ a. False
- ☒ b. True ✓



The correct answer is: True

The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

- ☒ a. True ✓
- ☐ b. False



الاجابة الصحيحة

The correct answer is: True

In the elastic collisions, the linear momentum is conserved, while the kinetic energy is not:

Select one:

- ☐ a. True
- ☒ b. False ✓



The correct answer is: False

The best equipment to measure the thickness of the paper is the Vernier caliper

Select one:

- ☐ a. True
- ☒ b. False ✓



The correct answer is: False

The sum of the change in kinetic and potential energy is always positive

Select one:

- ☐ a. True
- ☒ b. False ✓



The correct answer is: False

The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

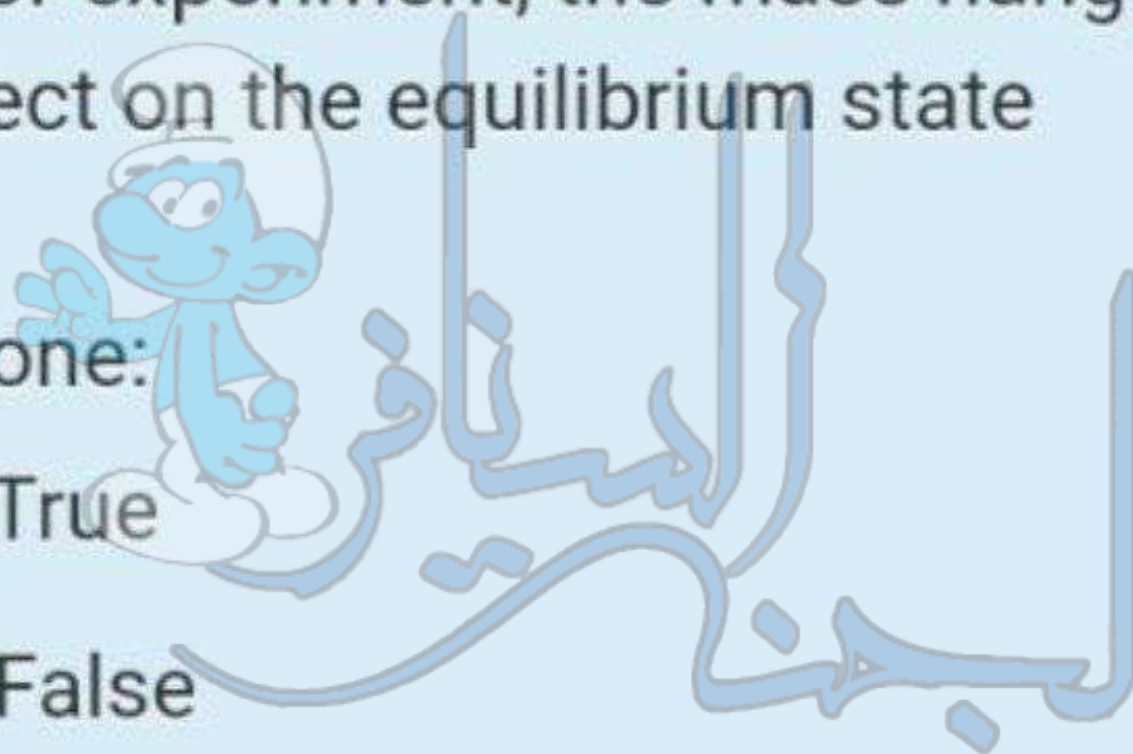
- ☐ a. False
- ☐ b. True



In vector experiment, the mass hangers do not affect on the equilibrium state

Select one:

- ☐ a. True
- ☐ b. False



مجموع التغير في الطاقة الحركية والكامنة إيجابي دائمًا

حدد واحدًا:

☐ أ. صحيح

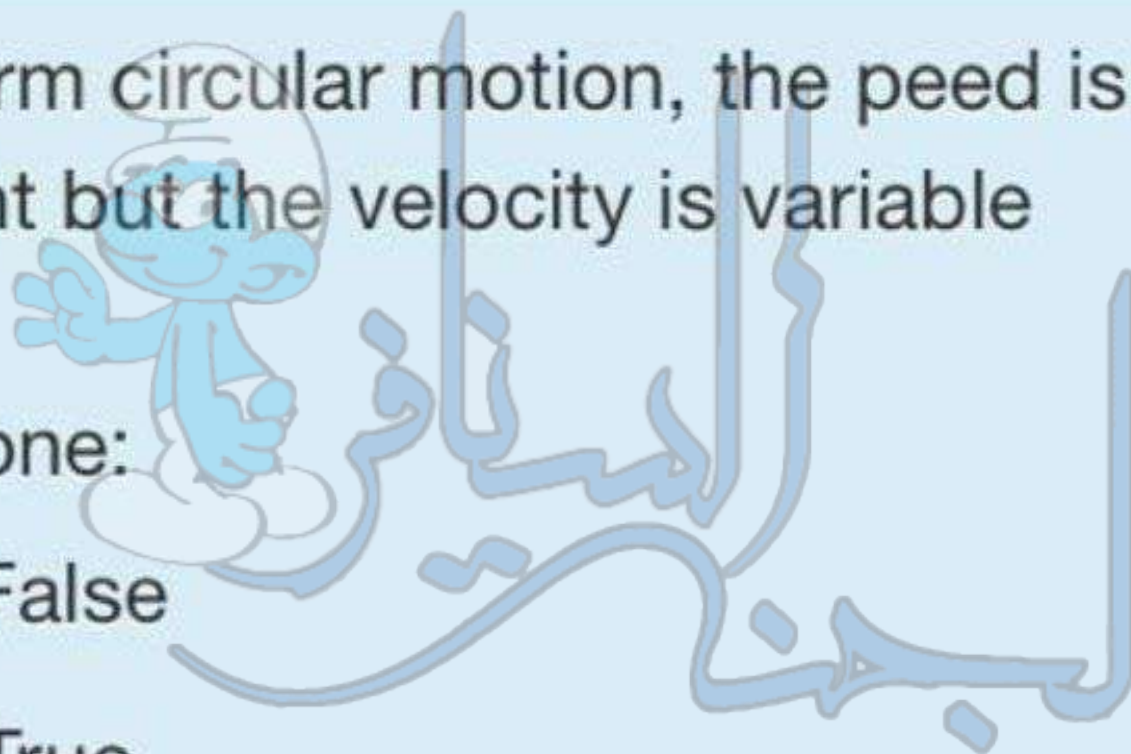
☐ ب. خاطئة



In uniform circular motion, the speed is constant but the velocity is variable

Select one:

- ☐ a. False
- ☐ b. True



It is wrong to say my weight 70 kg



اختر أحد الخيارات

صح



خطأ



The relationship between the length of the thread of the pendulum and the square of the periodic time

Select one:

- ☒ A. linear direct proportional
- ☐ B. curve relationship
- ☐ C. inverse linear proportional
- ☐ D. no relation

In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

Select one:

- ☐ a. The mass(M) will stop.
- ☐ b. The speed of the mass (M) will increase.
- ☐ c. The speed of the mass (M) will decrease
- ☐ d. The mass (M) will move in constant speed.

It is wrong to say my weight 70 kg

Select one:

☒ True

☐ False



In the inelastic collisions experiment, we can say

Select one:

- ☐ a. **The linear momentum always conserved ($P_f = P_i$).**
- ☐ b. **$V(m_1)_{\text{final}} = V(m_1)_{\text{initial}}$**
- ☐ c. **$(K.E_f) \neq (K.E_i)$**
- ☐ d. **All of them**

In the inelastic collisions experiment, we can say

Select one:

- ☐ a. **The linear momentum always conserved ($P_f = P_i$).**
- ☐ b. **$V(m_1)_{\text{final}} = V(m_1)_{\text{initial}}$**
- ☐ c. **$(K.E_f) \neq (K.E_i)$**
- ☐ d. **All of them**

The most suitable tool for measuring the thickness of a razor blade is vernier caliper



المستأف

اخترأحد الخيارات

صح



خطأ



In gravitational potential energy experiment,
always $\Delta(K.E) = \Delta(U.E)$



اختر أحد الخيارات

صح



خطأ



In gravitational potential energy experiment,
always $\Delta(K.E) = \Delta(U.E)$



اختر أحد الخيارات

صح



خطأ



In gravitational potential energy experiment, always
 $\Delta(K.E) + \Delta(U.E) = \text{zero}$



اختر أحد الخيارات

صح



خطأ



In gravitational potential energy experiment, always
 $\Delta(K.E) + \Delta(U.E) = \text{zero}$



اختر أحد الخيارات

صح



خطأ

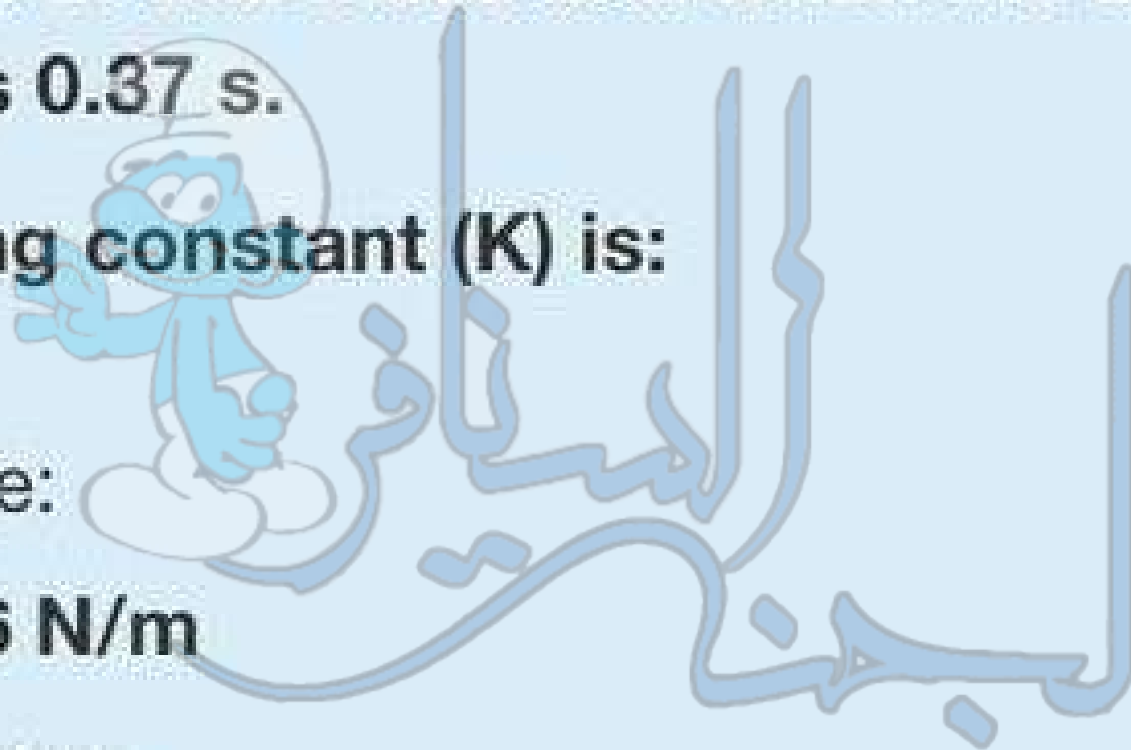


Suspend a body of mass (30 g) with a spring, if you know that the periodic time for the existence of this mass and when the elongation is fixed is 0.37 s.

The spring constant (K) is:

Select one:

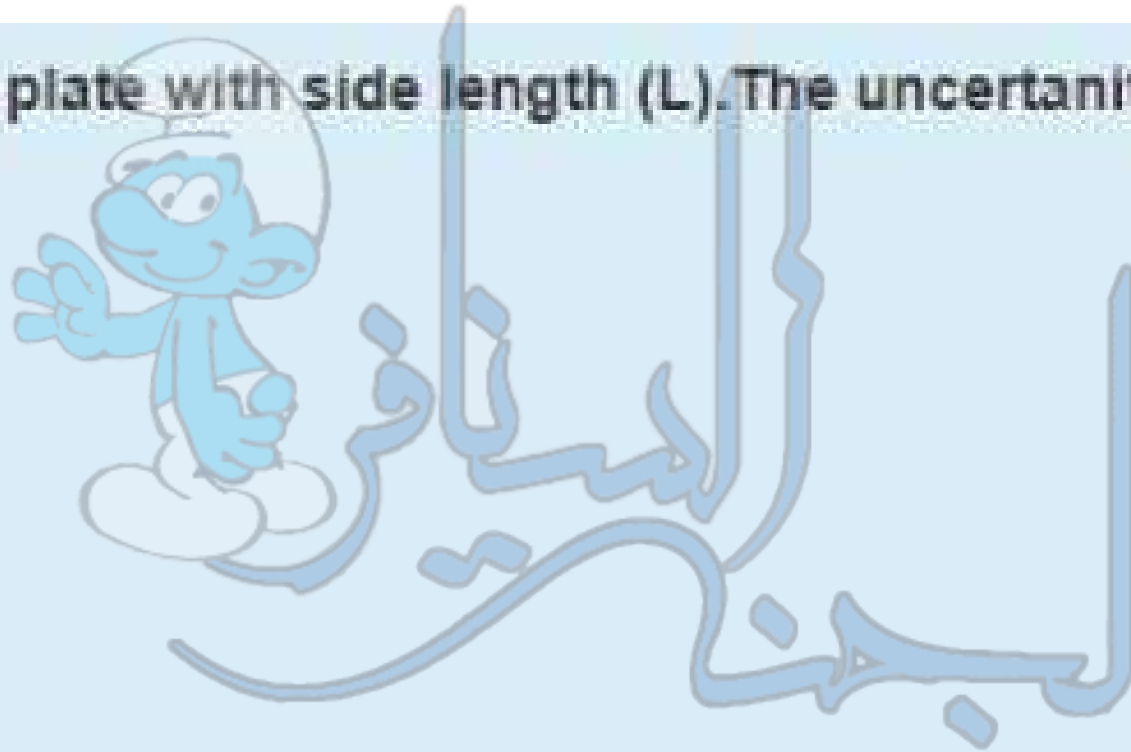
- ☐ a. 8.6 N/m
- ☐ b. 6 N/m
- ☐ c. 10 N/m
- ☐ d. 12 N/m



Square-shaped plate with side length (L). The uncertainty (error) in its area is:

Select one:

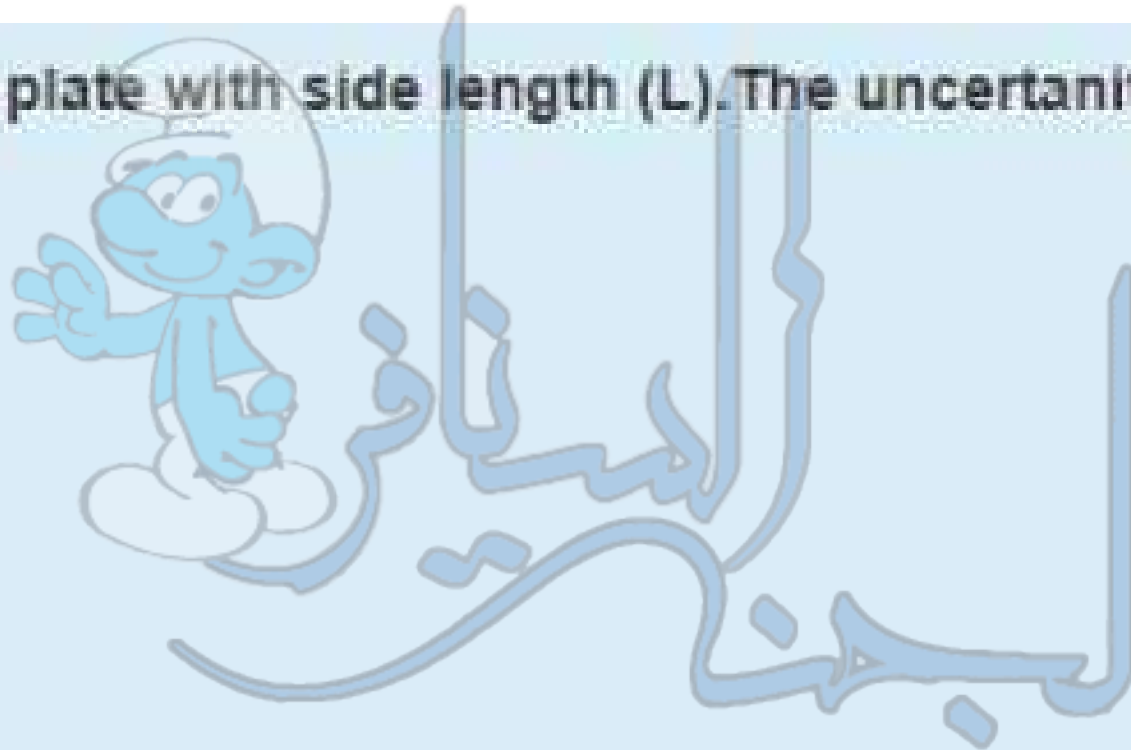
- ☐ a. $4L \cdot \Delta L$
- ☐ b. $4L^2 \cdot \Delta L$
- ☐ c. $2L^2 \cdot \Delta L$
- ☐ d. $2L \cdot \Delta L$



Square-shaped plate with side length (L). The uncertainty (error) in its area is:

Select one:

- ☐ a. $4L \cdot \Delta L$
- ☐ b. $4L^2 \cdot \Delta L$
- ☐ c. $2L^2 \cdot \Delta L$
- ☐ d. $2L \cdot \Delta L$



In gravitational potential energy experiment, always
 $\Delta(K.E) + \Delta(U.E) = \text{zero}$



اختر أحد الخيارات

صح



خطأ



In gravitational potential energy experiment, always
 $\Delta(K.E) + \Delta(U.E) = \text{zero}$



اختر أحد الخيارات

صح



خطأ



The most appropriate tool to measure the depth of water in a cup is vernnier caliper.

Select one:



True



False



The most appropriate tool to measure the depth of water in a cup is vernnier caliper.

Select one:



True



False



It is wrong to say my weight 70 N.

Select one:

☐ True

☐ False



It is wrong to say my weight 70 N.

Select one:

☐ True

☐ False



It is wrong to say my weight 70 N.

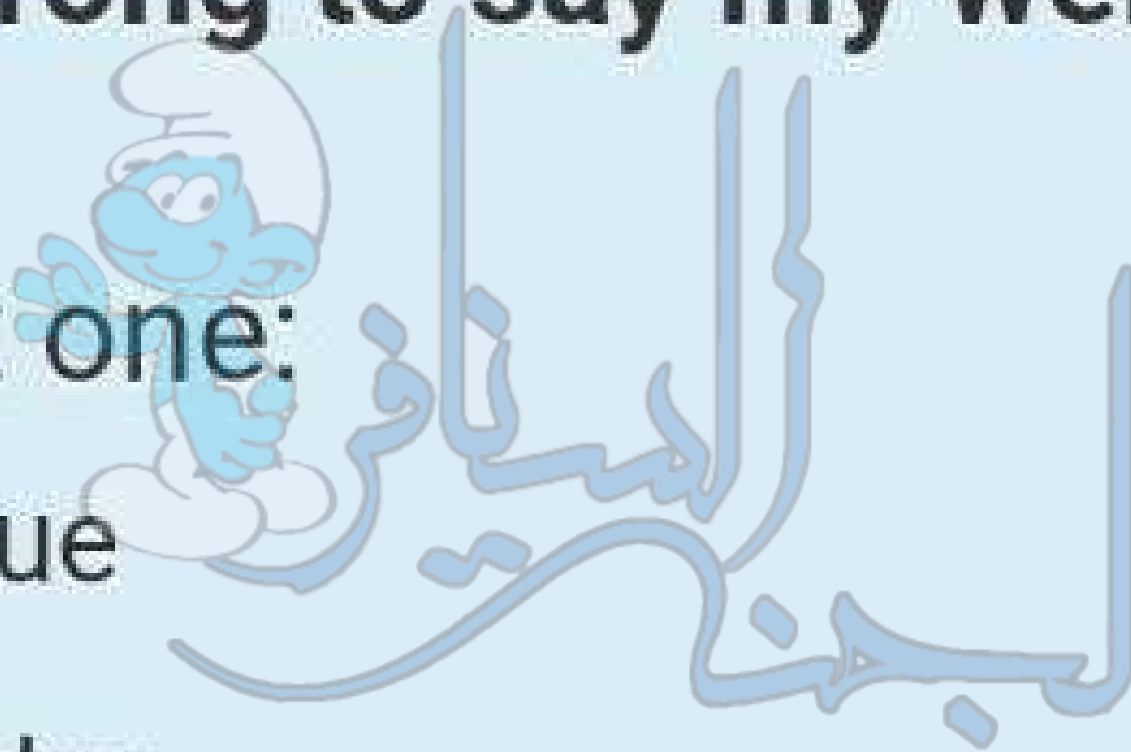
Select one:



True



False



It is wrong to say my weight 70 N.

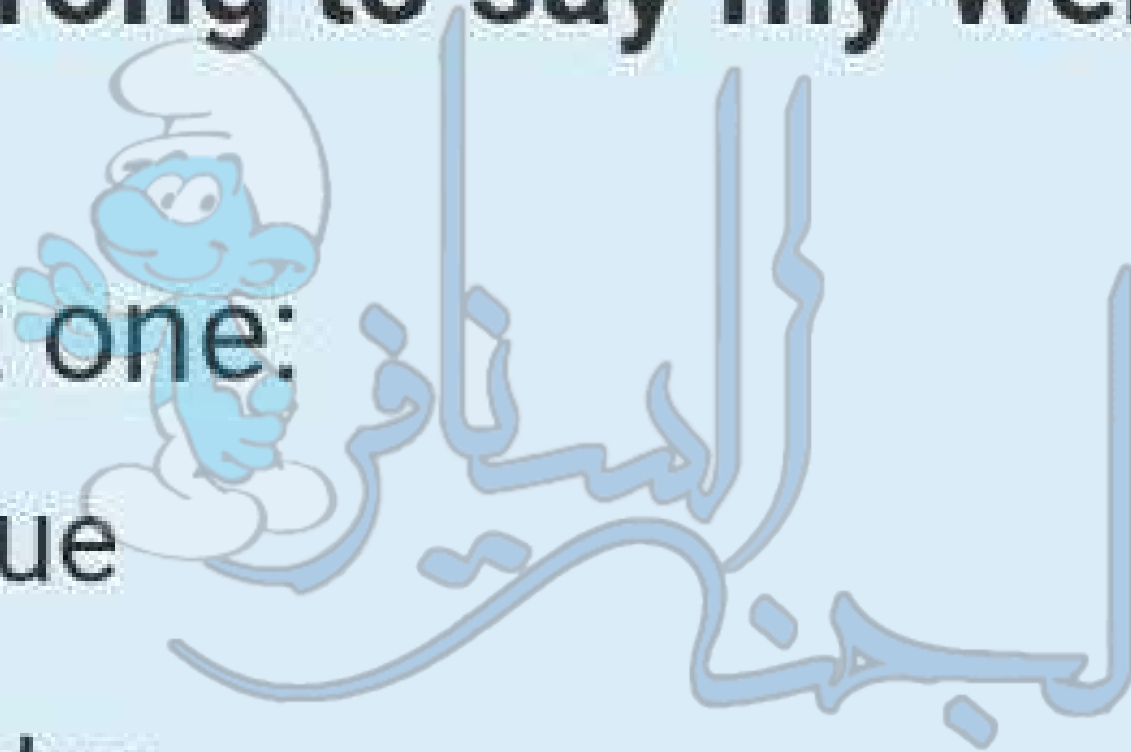
Select one:



True



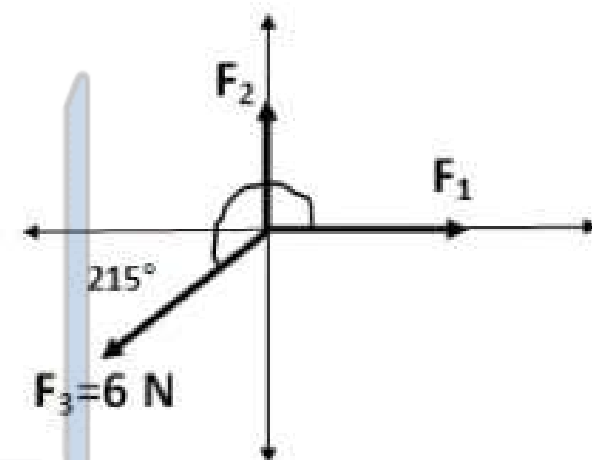
False



Three forces are in equilibrium, depending on the given figure. The value of (F_2) is

Select one:

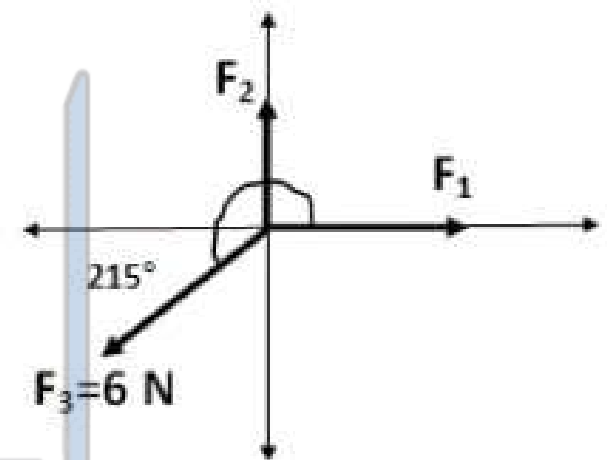
- ☐ a. 9.3 N
- ☐ b. 3.44 N
- ☐ c. 8.5 N
- ☐ d. 6 N



Three forces are in equilibrium, depending on the given figure. The value of (F_2) is

Select one:

- ☐ a. 9.3 N
- ☐ b. 3.44 N
- ☐ c. 8.5 N
- ☐ d. 6 N



In a gravitational potential energy experiment,
(when (m) reaches Earth, the mass (M

اختر أحد الخيارات

- a. The speed of the mass (M) will decrease ☐
- .b. The mass(M) will stop ☐
- .c. The mass (M) will move in constant speed ☐
- .d. The speed of the mass (M) will increase ☐

In a gravitational potential energy experiment,
(when (m) reaches Earth, the mass (M

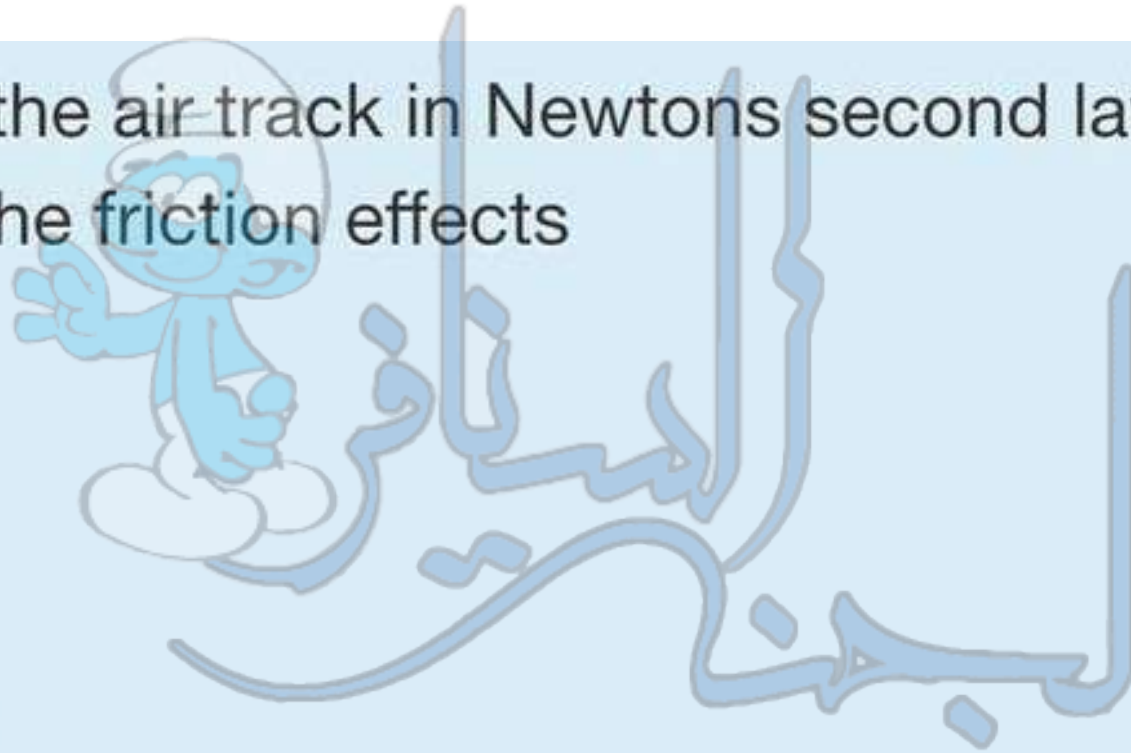
اختر أحد الخيارات

- a. The speed of the mass (M) will decrease ☐
- b. The mass(M) will stop ☐
- c. The mass (M) will move in constant speed ☐
- d. The speed of the mass (M) will increase ☐

The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

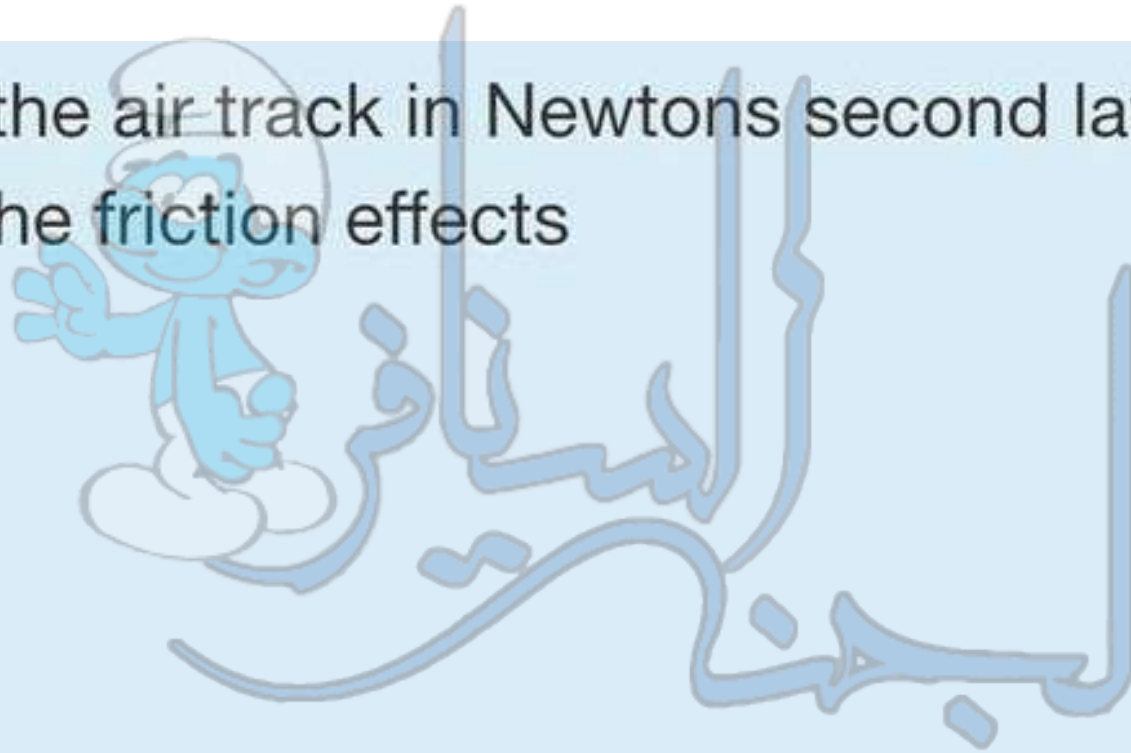
- ☐ a. True
- ☐ b. False



The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

- ☐ a. True
- ☐ b. False



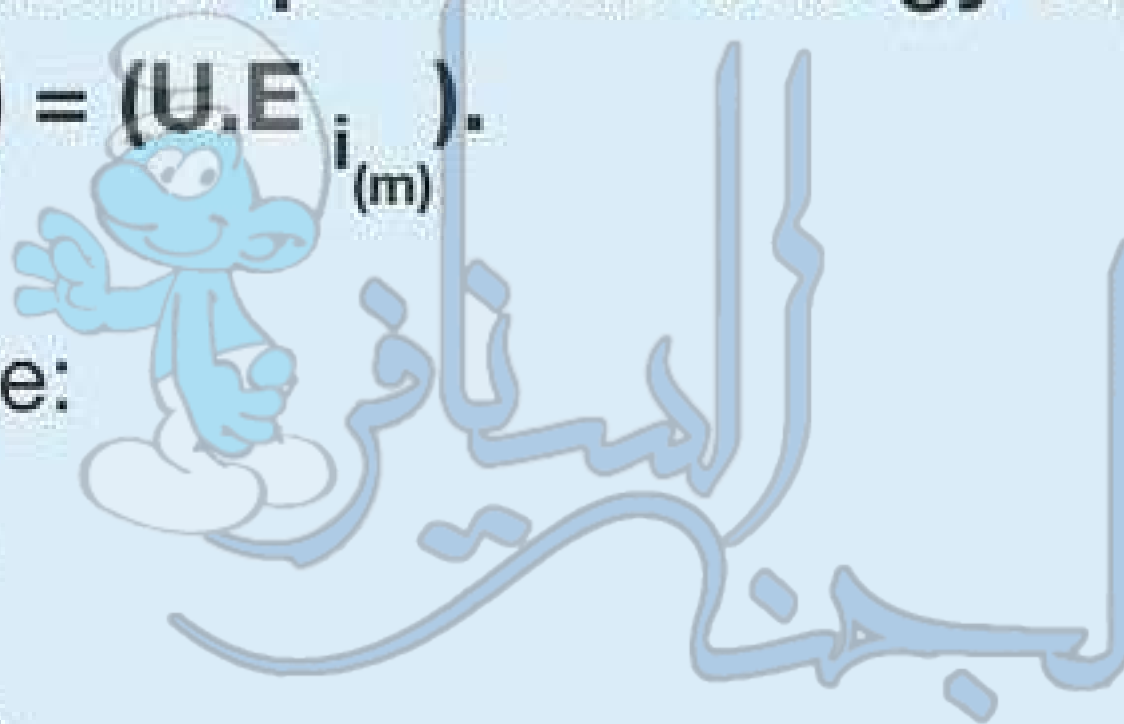
In gravitational potential energy experiment,

$$(K.E_{f_{(M,m)}}) = (U.E_{i_{(m)}}).$$

Select one:

☐ True

☐ False



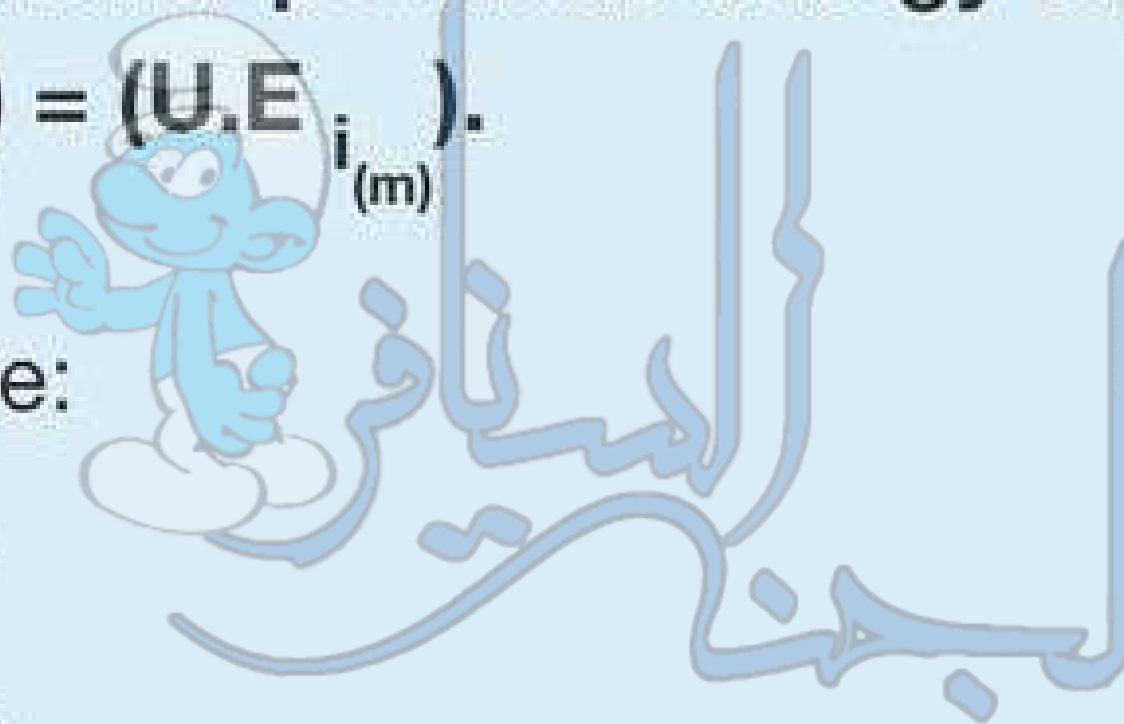
In gravitational potential energy experiment,

$$(K.E_{f_{(M,m)}}) = (U.E_{i_{(m)}}).$$

Select one:

☐ True

☐ False



The unit of spring constant is (kg/s^2) .

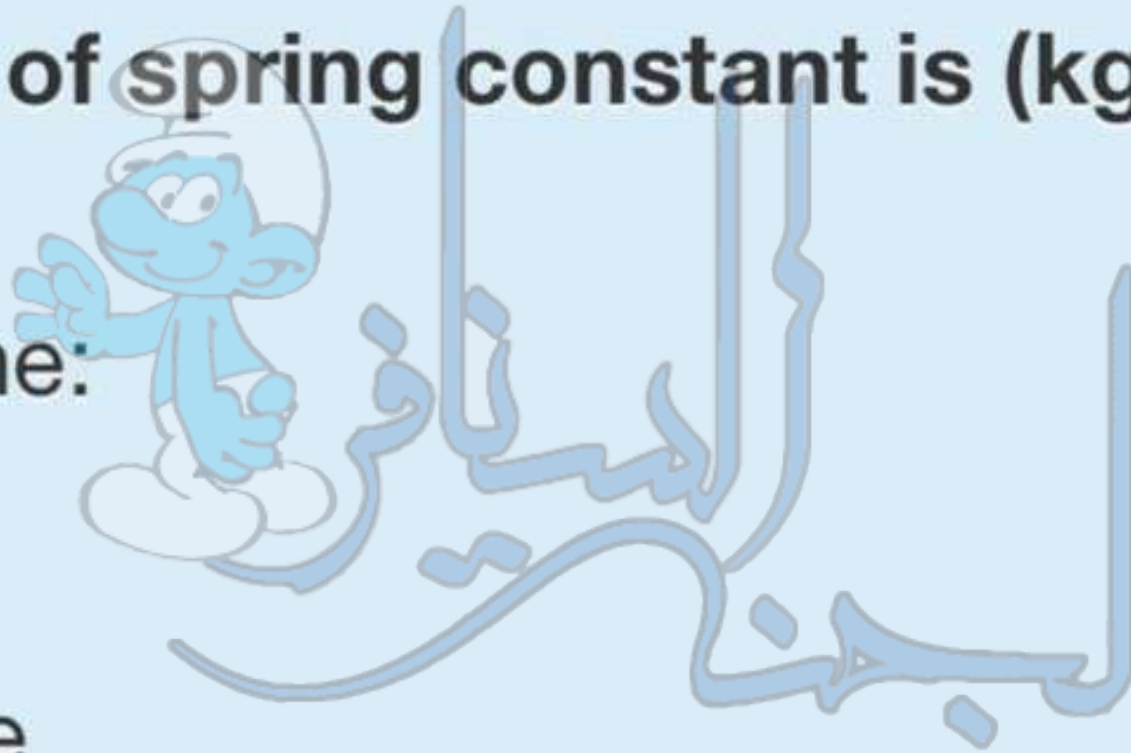
Select one:



True



False



The unit of spring constant is (kg/s^2) .

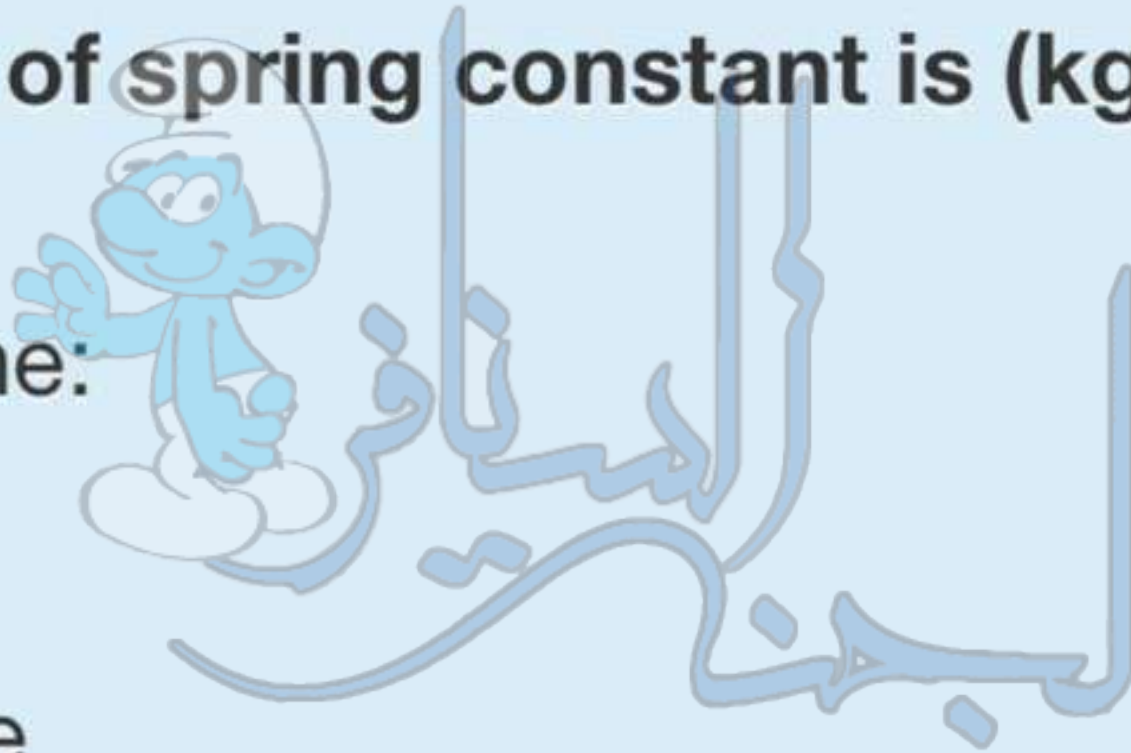
Select one:



True



False



The number of significant digits in this number (0.00280×10^5) is 3 (s.f)

Select one:



True



False



الجامعة
الاسلامية
بجدة

The number of significant digits in this number (0.00280×10^5) is 3 (s.f)

Select one:



True



False



الجامعة
الاسلامية
بجدة

: we can say about (μ_s) and (μ_k) that



اختر أحد الخيارات

(A. $(\mu_s) < (\mu_k)$ ☐

(B. $(\mu_s) > (\mu_k)$ ☐

C. $-1 \geq (\mu_s) \text{ AND } (\mu_k) \geq 1$ ☐

D. $(\mu_s) \text{ AND } (\mu_k) \geq 1$ ☐

: we can say about (μ_s) and (μ_k) that



اختر أحد الخيارات

(A. $(\mu_s) < (\mu_k)$ ☐

(B. $(\mu_s) > (\mu_k)$ ☐

C. $-1 \geq (\mu_s) \text{ AND } (\mu_k) \geq 1$ ☐

D. $(\mu_s) \text{ AND } (\mu_k) \geq 1$ ☐

A ball mass (m), and radius (r).The uncertainty (error) of its volume

Select one:

- ☐ a. $8\pi.r^3.\Delta r$
- ☐ b. $8\pi.r.m.\Delta r$
- ☐ c. $8\pi.r^2.\Delta r$
- ☐ d. $4\pi.r^2.\Delta r$

A ball mass (m), and radius (r).The uncertainty (error) of its volume

Select one:

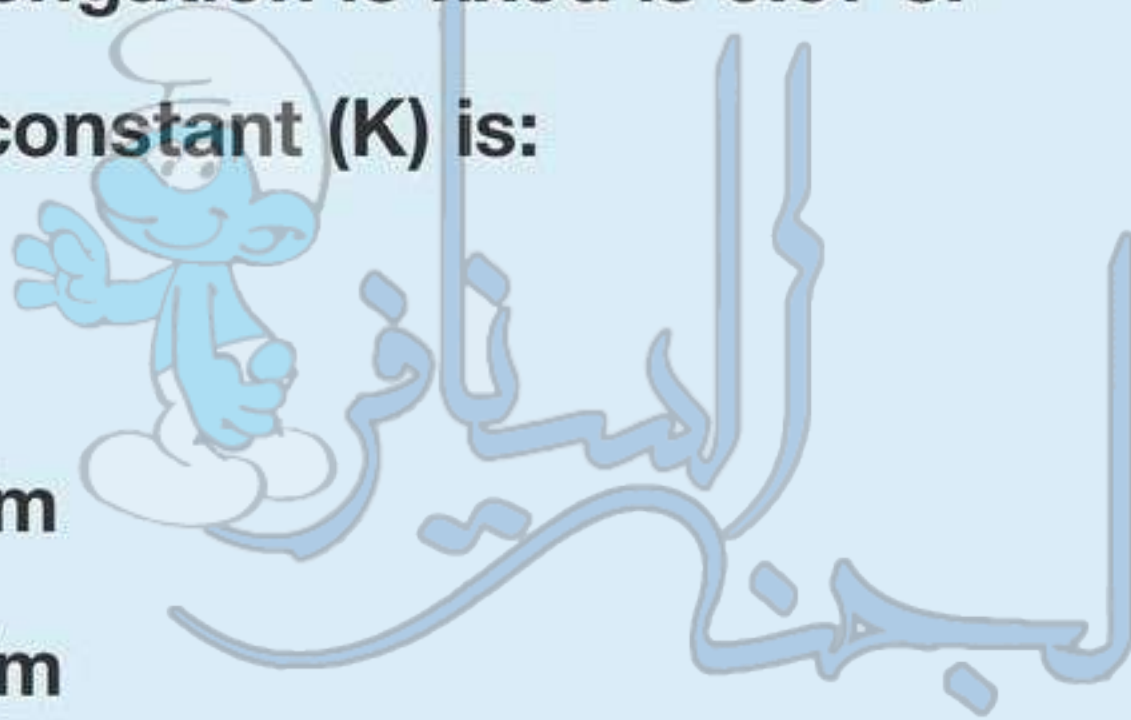
- ☐ a. $8\pi.r^3.\Delta r$
- ☐ b. $8\pi.r.m.\Delta r$
- ☐ c. $8\pi.r^2.\Delta r$
- ☐ d. $4\pi.r^2.\Delta r$

Suspend a body of mass (30 g) with a spring, if you know that the periodic time for the existence of this mass and when the elongation is fixed is 0.37 s.

The spring constant (K) is:

Select one:

- ☐ a. **10 N/m**
- ☐ b. **12 N/m**
- ☐ c. **6 N/m**
- ☐ d. **8.6 N/m**

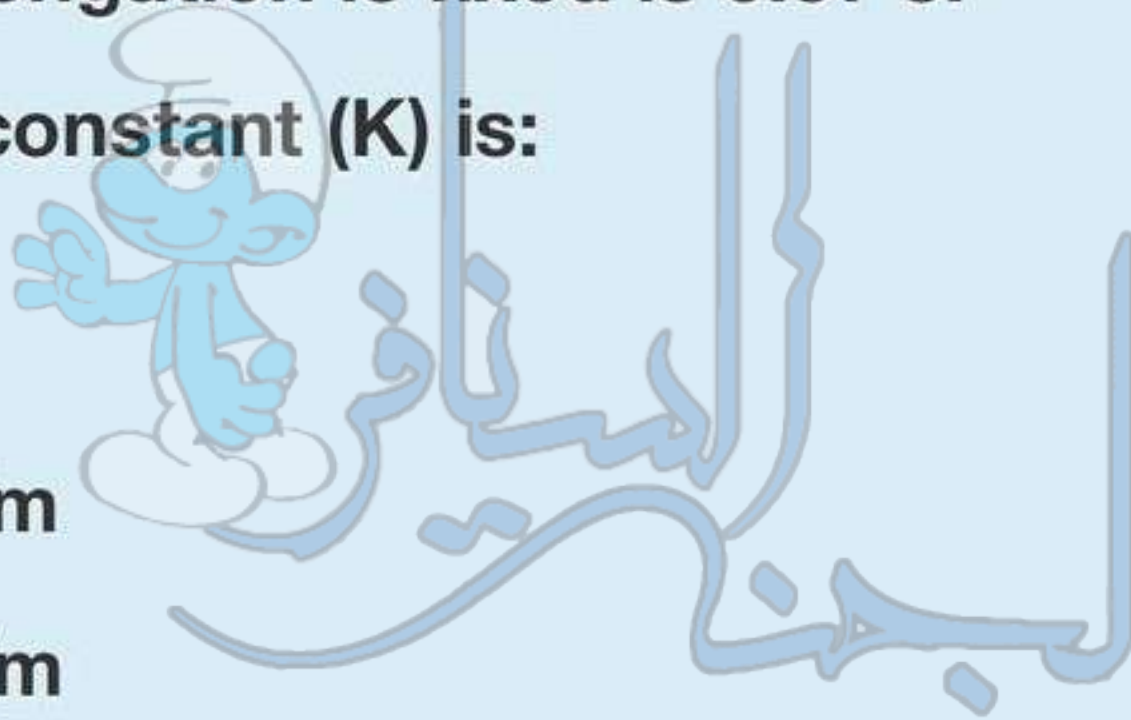


Suspend a body of mass (30 g) with a spring, if you know that the periodic time for the existence of this mass and when the elongation is fixed is 0.37 s.

The spring constant (K) is:

Select one:

- ☐ a. **10 N/m**
- ☐ b. **12 N/m**
- ☐ c. **6 N/m**
- ☐ d. **8.6 N/m**



Using the rules of significant numbers ($3.025 + 2.4 = 5.4$)

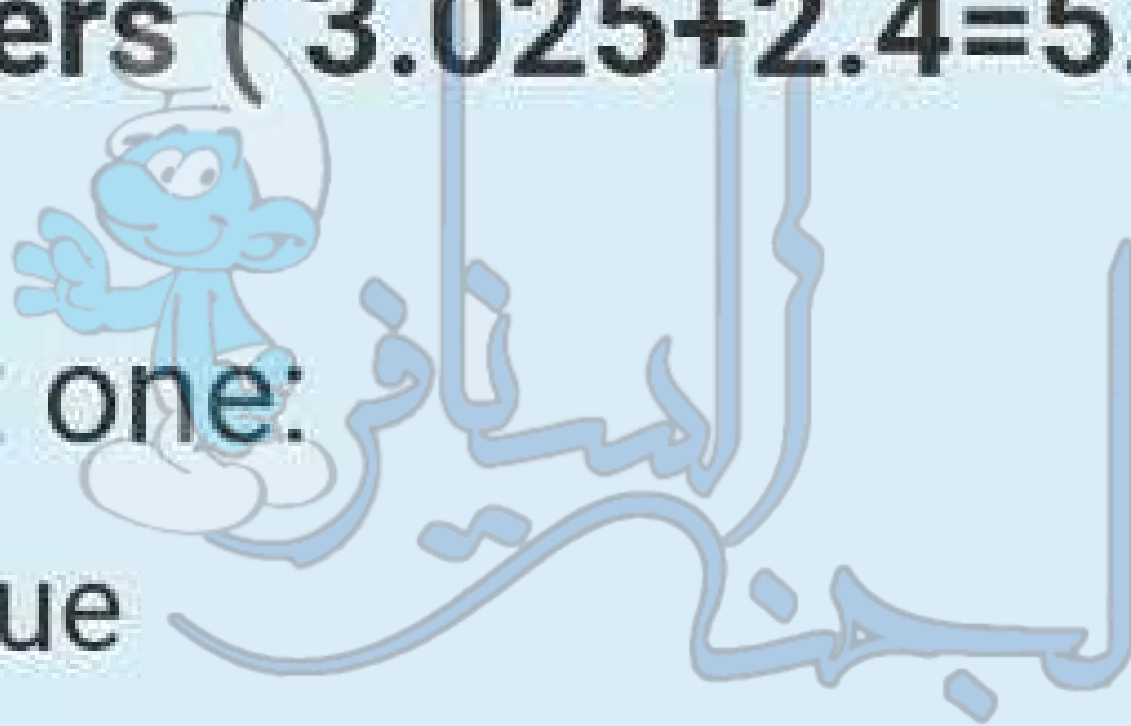
Select one:



True



False



It is wrong to say my weight 70 N.

Select one:



True



False



الاجابة الصحيحة

It is wrong to say my weight 70 N.

Select one:



True



False

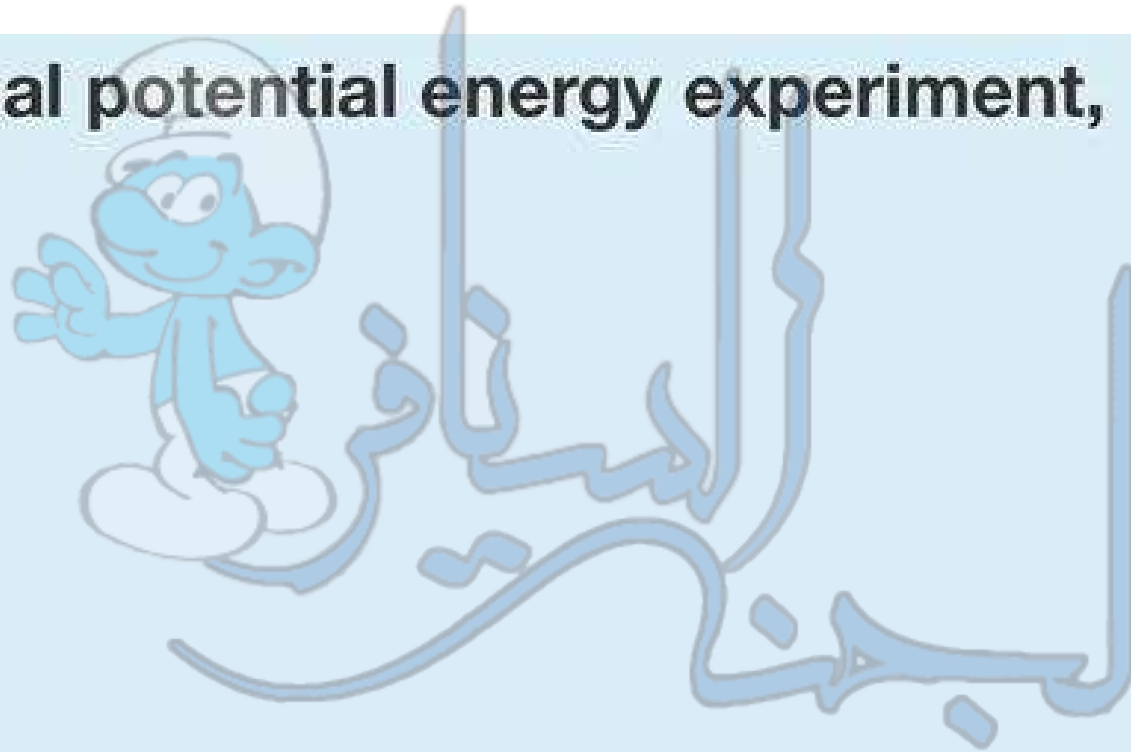


السلامة

In gravitational potential energy experiment, $(K.E_{f(M,m)}) = (U.E_{i(m)})$

Select one:

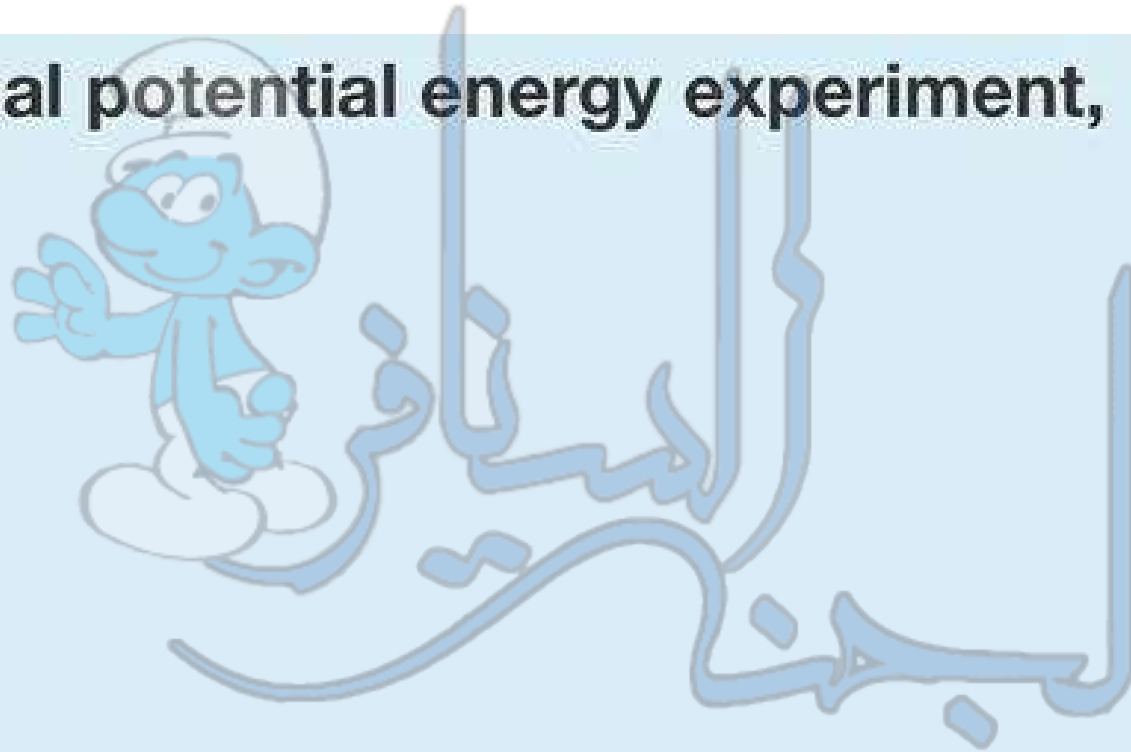
- ☐ True
- ☐ False



In gravitational potential energy experiment, $(K.E_{f(M,m)}) = (U.E_{i(m)})$

Select one:

- ☐ True
- ☐ False



Square-shaped plate with side length (L).The uncertainty (error) in it area is:

Select one:

- ☐ a. $4L^2.\Delta L$
- ☐ b. $4L.\Delta L$
- ☐ c. $2L^2.\Delta L$
- ☐ d. $2L.\Delta L$



Square-shaped plate with side length (L).The uncertainty (error) in it area is:

Select one:

- ☐ a. $4L^2.\Delta L$
- ☐ b. $4L.\Delta L$
- ☐ c. $2L^2.\Delta L$
- ☐ d. $2L.\Delta L$



Using the rules of significant numbers (3.025 -
(2.4=0.6



المستأفرون
لجنة

اخترأحد الخيارات

صح



خطأ



The most suitable tool for measuring the thickness of a razor blade is vernnier caliper.

Select one:

☐ True

☐ False



The using of the air track in Newtons second law experiment is to decrease the friction effects



اخترأ أحد الخيارات

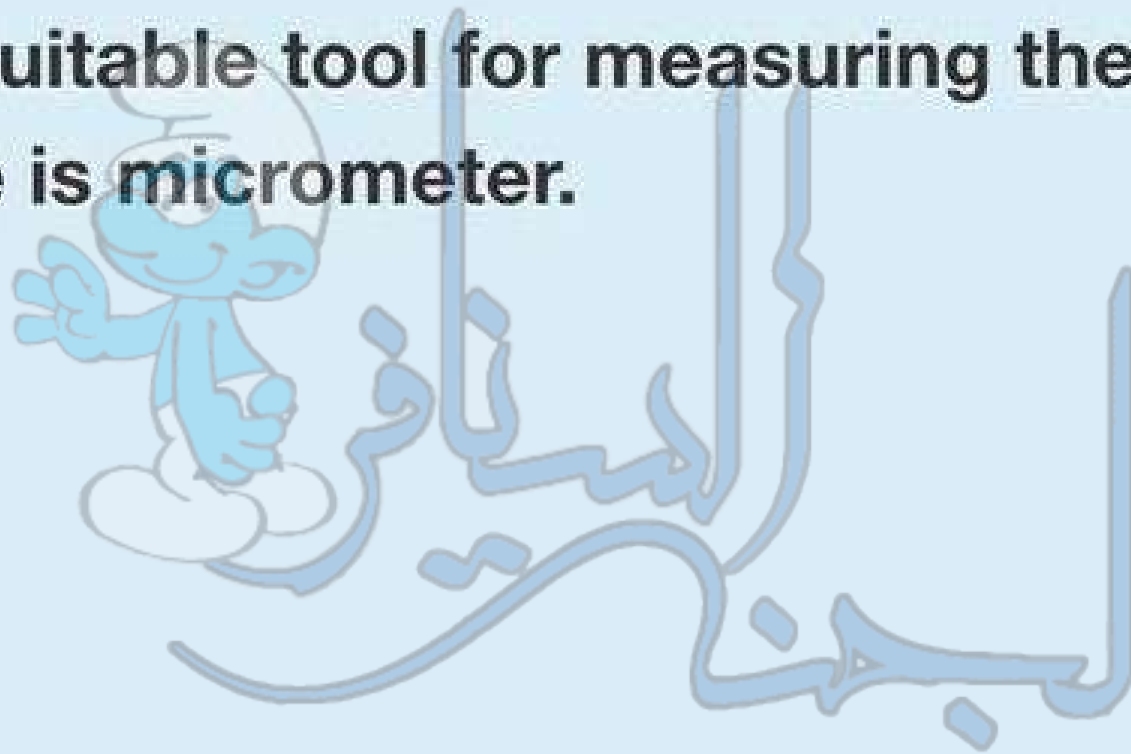
a. False ☐

b. True ☐

The most suitable tool for measuring the thickness of a razor blade is micrometer.

Select one:

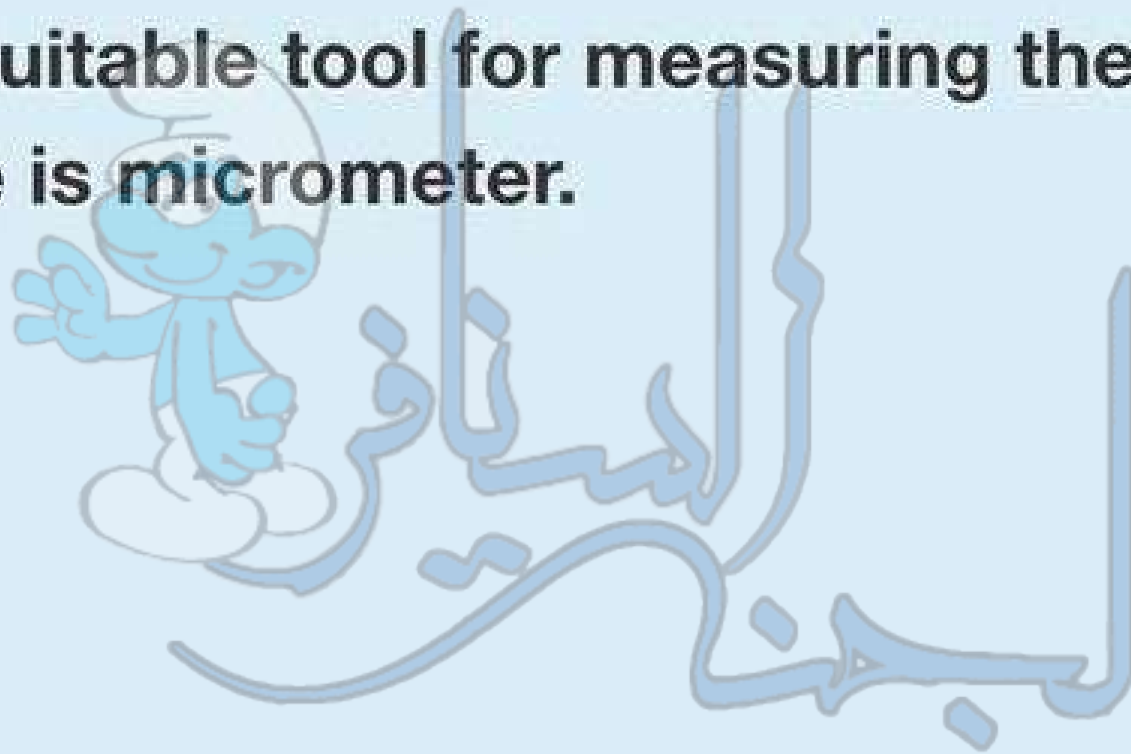
- ☐ True
- ☐ False



The most suitable tool for measuring the thickness of a razor blade is micrometer.

Select one:

- ☐ True
- ☐ False



It is wrong to say my weight 70 N.

Select one:



True



False



لجنة الامتحان

It is wrong to say my weight 70 N.

Select one:



True

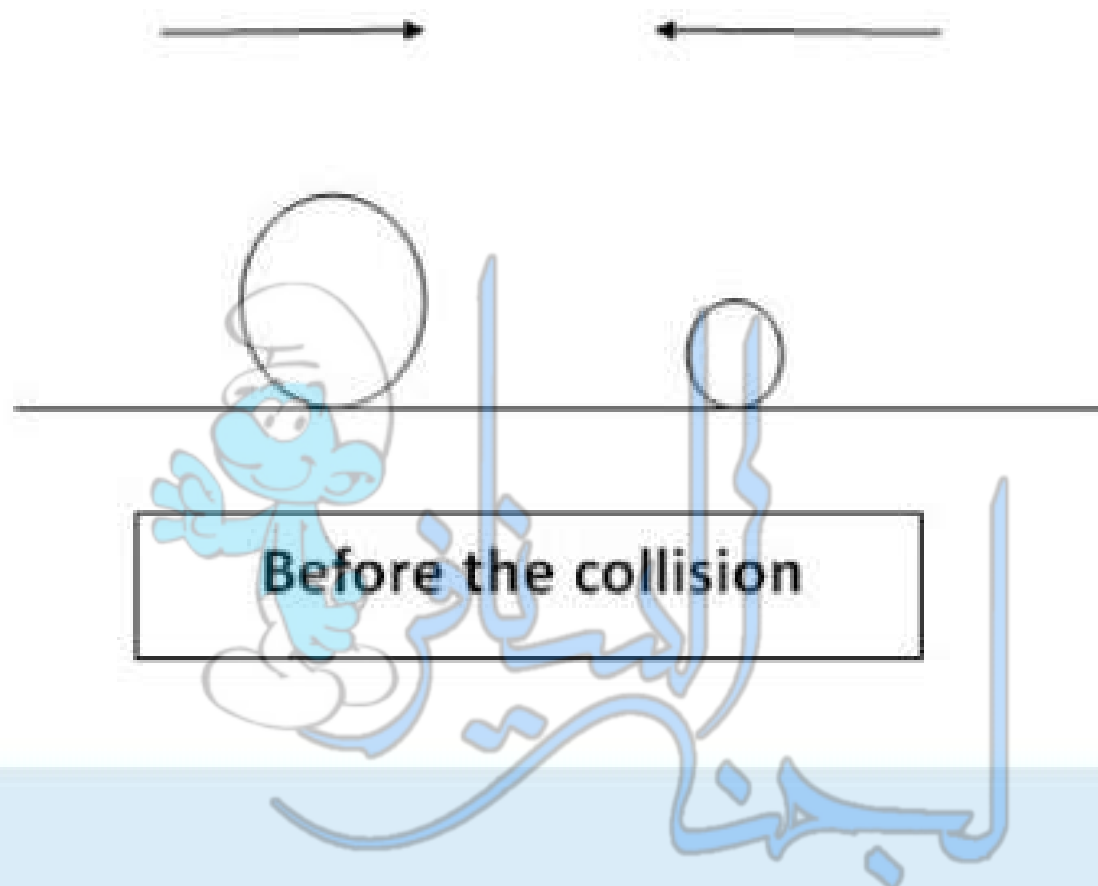


False



لجنة المسابقة

In the elastic collisions, if $(M=3m)$ and $V_{1i(M)} = 2 \text{ m/s}$, $V_{2i(m)} = 3 \text{ m/s}$, as the final linear momentum ($P_f = 18 \text{ kg.m/s}$). then (M) is:



Select one:

- ☐ a. 12 kg
- ☐ b. 4 kg
- ☐ c. 18 kg
- ☐ d. 6 kg

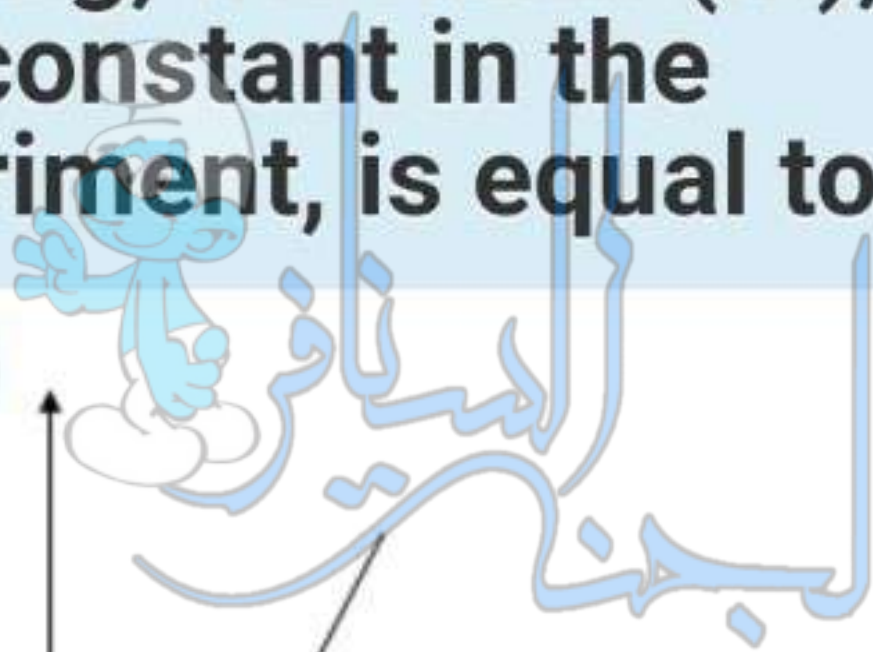
In the experiment of uniform circular motion, the student obtained the shown drawing, when the radius was fixed at (40 cm).

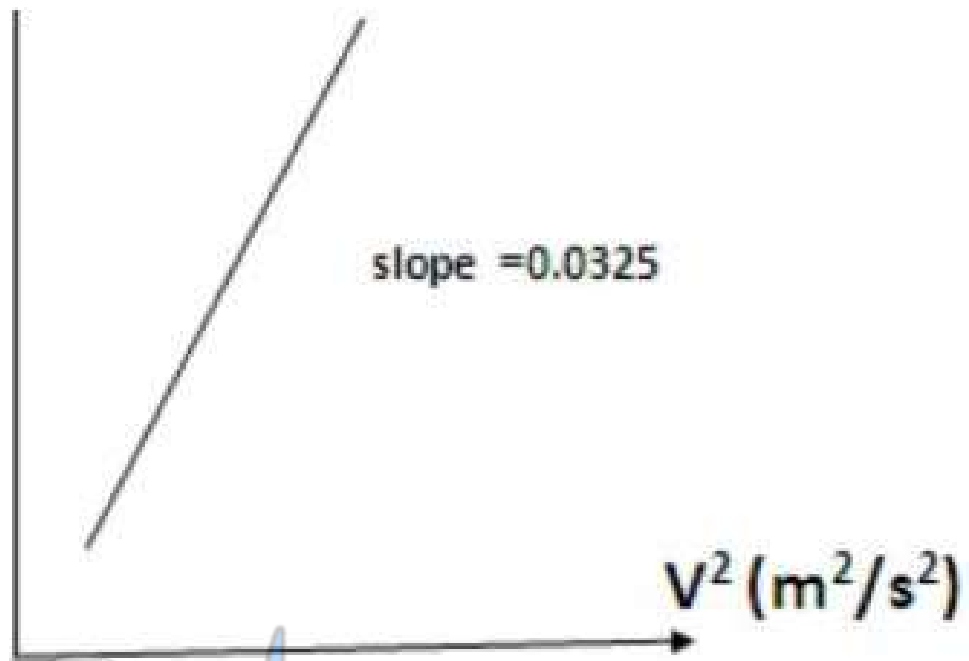
Based on the student's drawing, the mass (m), which was constant in the experiment, is equal to:

f_c (N)

slope = 0.0325

v^2 (m^2/s^2)





Select one:

- ☐ a. 13 g
- ☐ b. 10 g
- ☐ c. 12 g
- ☐ d. 9 g

In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

Select one:

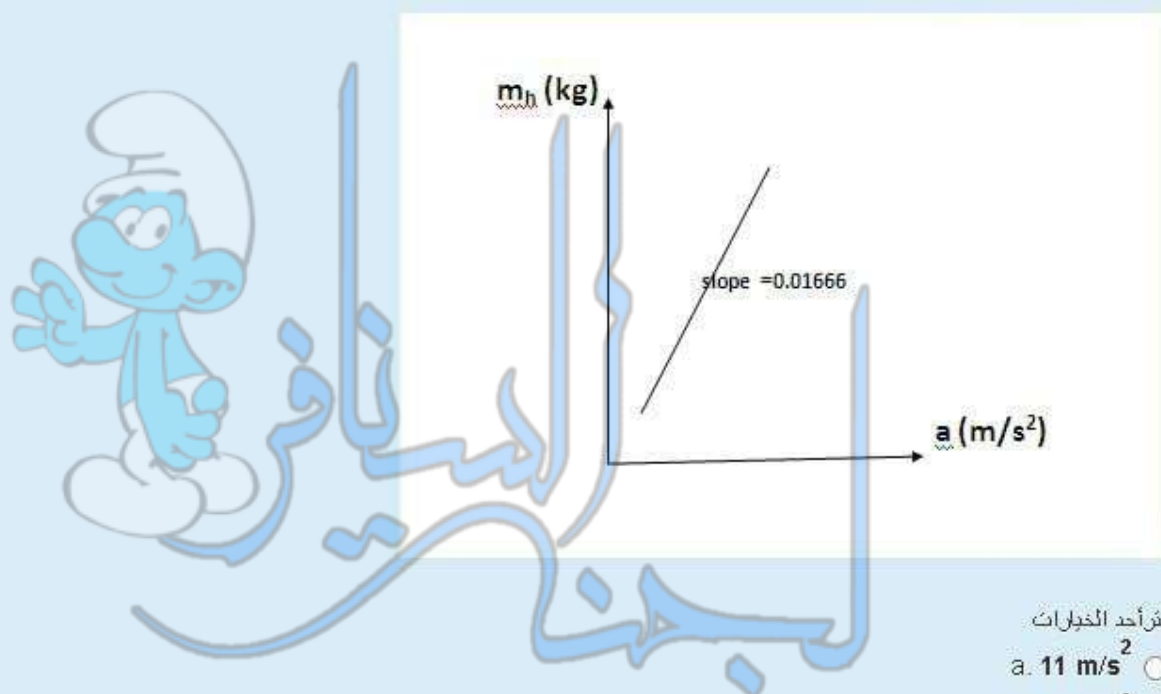
- ☐ a. The speed of the mass (M) will increase.
- ☐ b. The mass(M) will stop.
- ☐ c. The mass (M) will move in constant speed.
- ☐ d. The speed of the mass (M) will decrease

In Newton's second law experiment, when ($m_h = 15\text{g}$) and the total mass was (175 g). The acceleration of (m_h) is

Select one:

- ☐ a. 2 m/s^2
- ☐ b. 0.86 m/s^2
- ☐ c. 1 m/s^2
- ☐ d. 0.125 m/s^2

In Newton's second law experiment, a student obtained the graph shown, when the total mass was constant (200g). Depending on his drawing, the magnitude of the acceleration of the Earth's gravitational pull



اختر أحد الخيارات

- a. 11 m/s^2 ☐
- b. 9 m/s^2 ☐
- c. 8 m/s^2 ☐
- d. 12 m/s^2 ☒

The relation between the mass of pendulum (m), and the periodic time (one oscillation time) (T) at the same length of pendulum thread

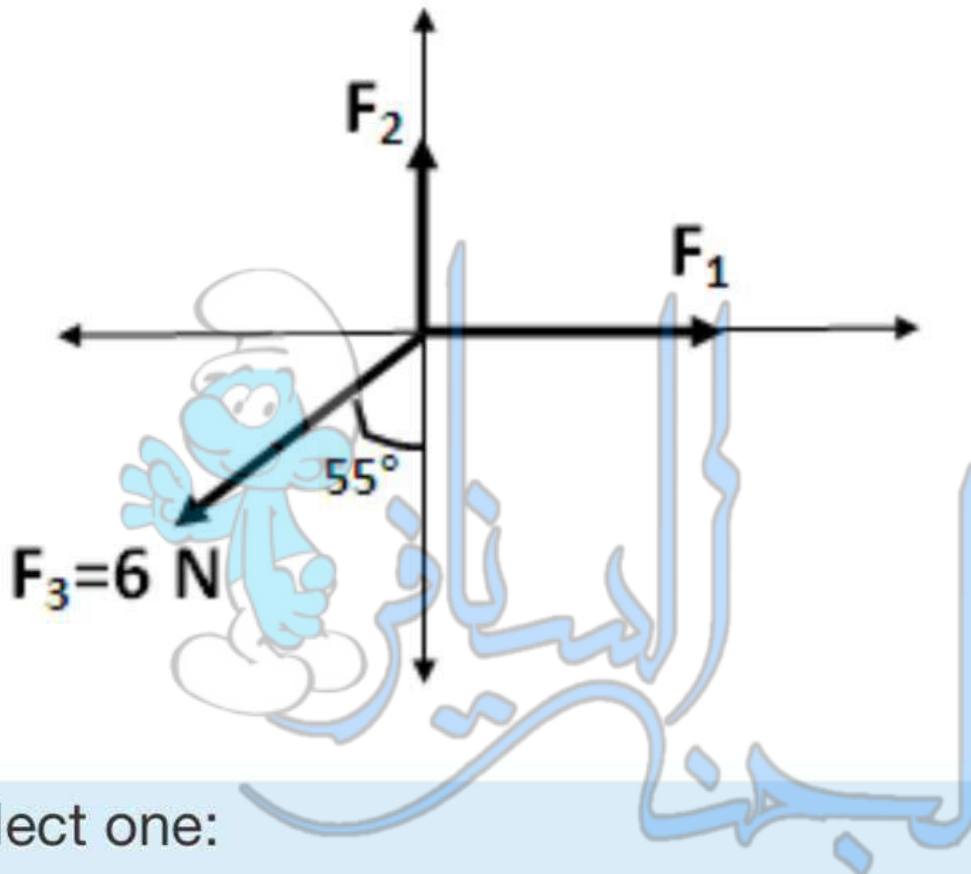


الأسئلة المتكررة

اختر أحد الخيارات

- A. linear direct ☐
- B. curve relationship ☒
- C. inverse linear ☐
- D. no relation ☐

Three forces are in equilibrium, depending on the given figure. The value of (F_2) is



Select one:

- ☐ a. 8.5 N
- ☒ b. 3.44 N
- ☐ c. 6 N
- ☐ d. 9.3 N

In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

Select one:

- ☐ a. The speed of the mass (M) will decrease
- ☐ b. The mass (M) will move in constant speed.
- ☐ c. The mass(M) will stop.
- ☐ d. The speed of the mass (M) will increase.

Square-shaped plate with side length (L). The uncertainty (error) in it area is:

Select one:

- ☐ a. $4L \cdot \Delta L$
- ☐ b. $2L \cdot \Delta L$
- ☐ c. $4L^2 \cdot \Delta L$
- ☐ d. $2L^2 \cdot \Delta L$

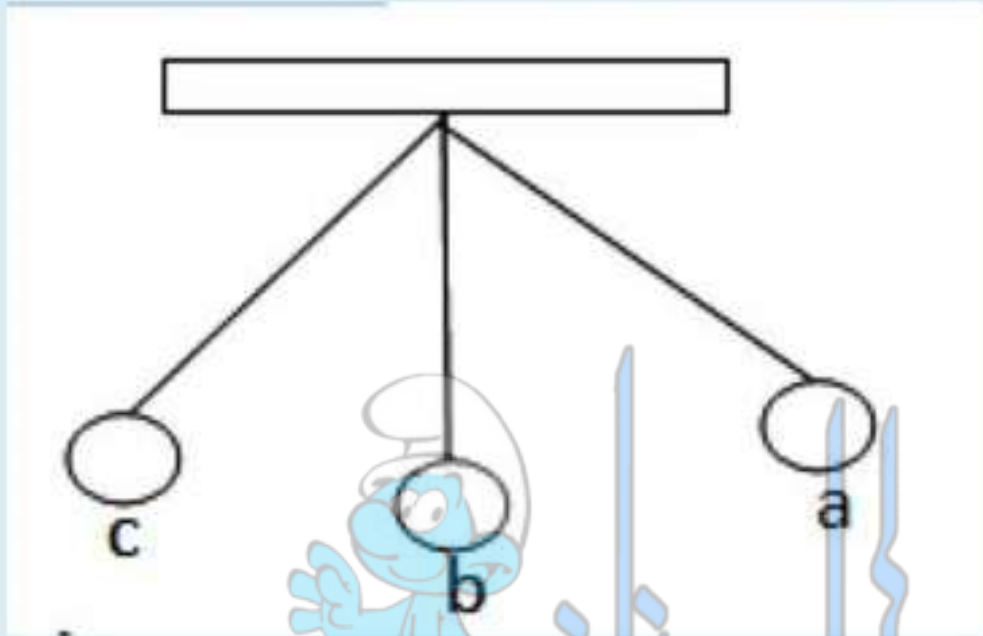


In Uniform circular motion experiment, the benefit of using crocodile tongs, is

Select one:

- ☐ a. To keep ($Mg = f_c$).
- ☐ b. To keep the radius constant.
- ☐ c. To keep the magnitude of the velocity constant.
- ☐ d. All of them

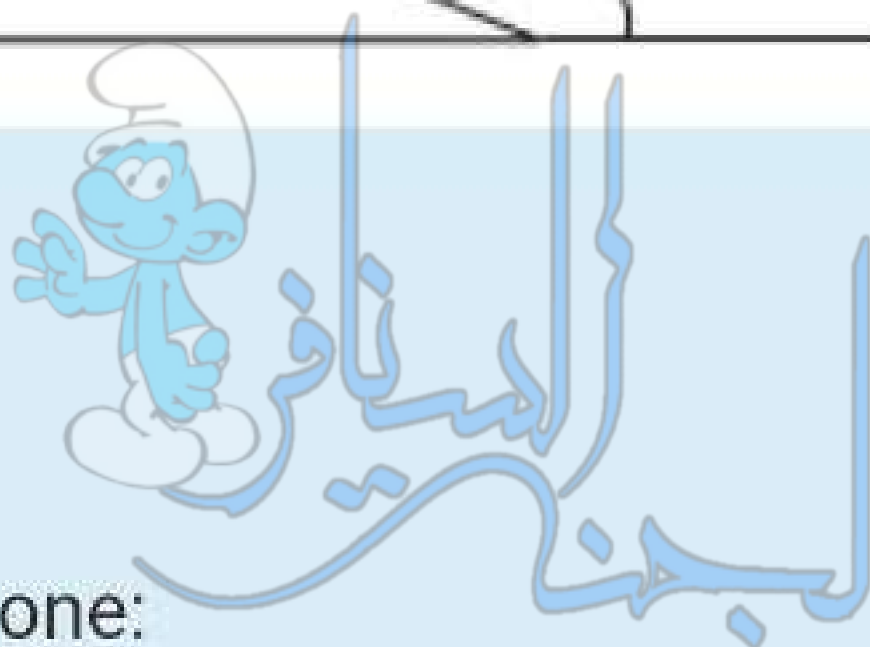
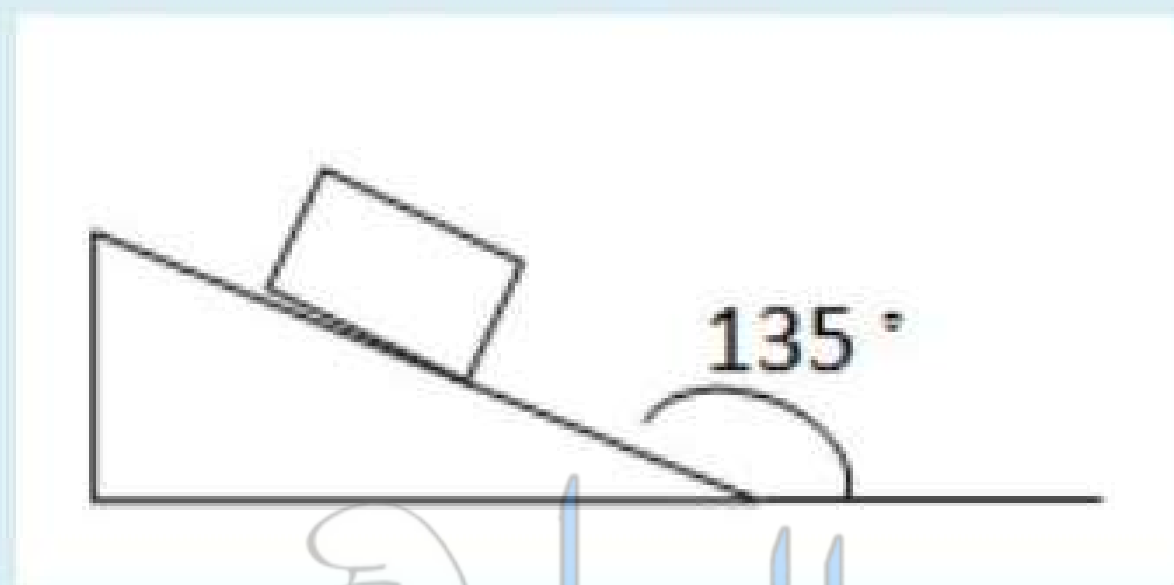
If pendulum take time (0.067s) to move from (a) to (C), then the period of one oscillation (T) :



Select one:

- ☐ A. 0.134 S
- ☐ B. 0.67 S
- ☐ C. 0.067 S
- ☐ D. 0.268 S

(m) is about to slide Over the inclined level shown. so (μ_s):



Select one:

- ☐ A. 1
- ☐ B. 0.6
- ☐ C. -1
- ☐ D. 0.4

In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

Select one:

- ☐ a. The speed of the mass (M) will decrease
- ☐ b. The mass(M) will stop.
- ☐ c. The speed of the mass (M) will increase.
- ☐ d. The mass (M) will move in constant speed.

The relation between the mass of pendulum (m), and the periodic time (one oscillation time) (T) at the same length of pendulum thread



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- A. inverse linear ☐
- B. linear direct ☐
- C. curve relationship ☒
- D. no relation ☐

201111-PHYSICS LAB. 1-9

▶ 2021/2020-1-30201111-PHYSICS LAB. 1-9 ▶ General ▶

on 8

answered

out of 2.00

question

The relationship between the length of the thread of the pendulum and the square of the periodic time

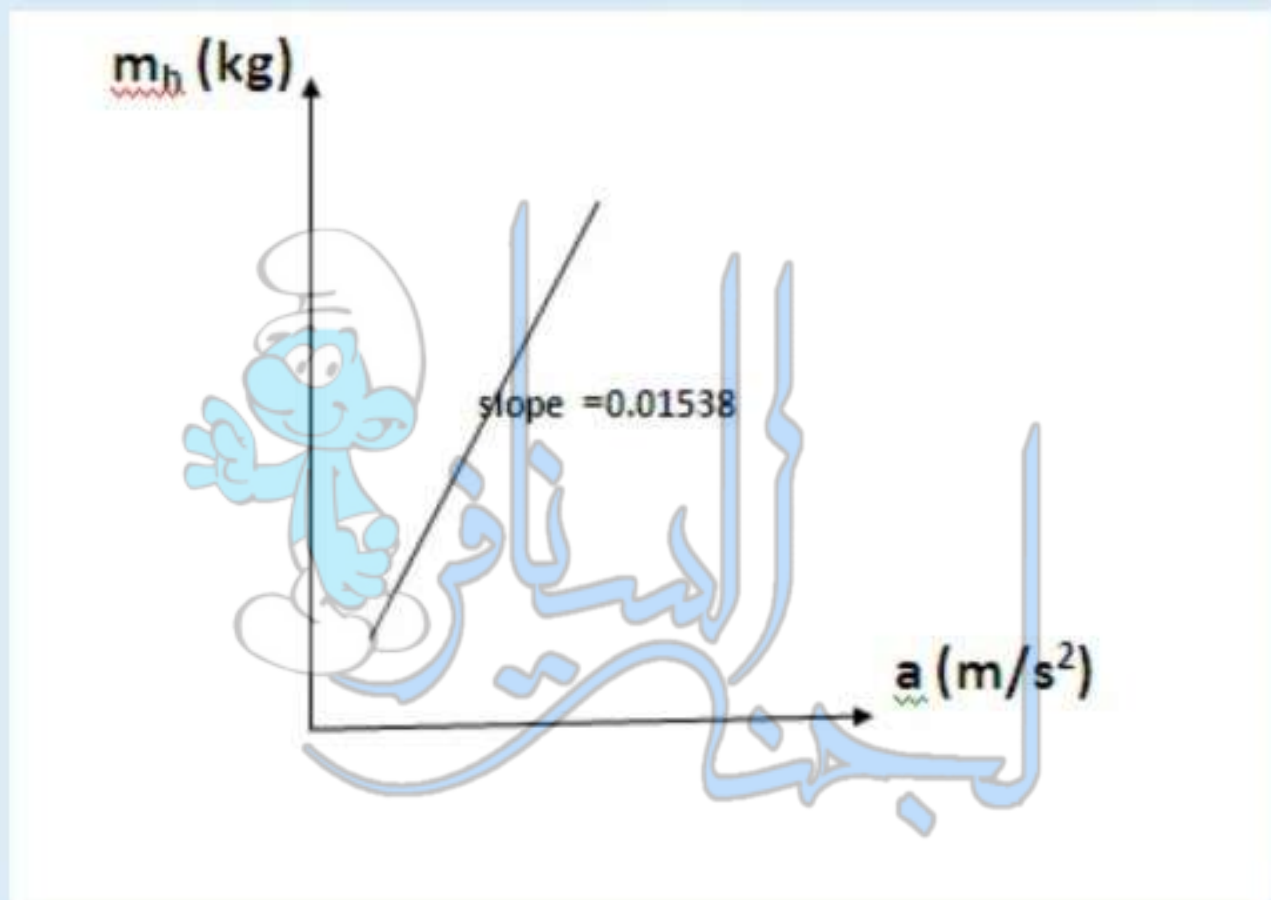
Select one:

- ☐ A. linear direct proportional
- ☐ B. curve relationship
- ☐ C. inverse linear proportional
- ☐ D. no relation

Next page

In Newton's second law experiment, a student obtained the graph shown, when the total mass was constant (200g).

Depending on his drawing, the magnitude of the acceleration of the Earth's gravitational pull



اختر أحد الخيارات

a. 12 m/s^2 ☐

b. 11 m/s^2 ☐

c. 9 m/s^2 ☐

d. 13 m/s^2 ☐

Suspend a body of mass (30 g) with a spring, if you know that the periodic time for the existence of this mass and when the elongation is fixed is 0.37 s

The spring constant (K) is



اختر أحد الخيارات

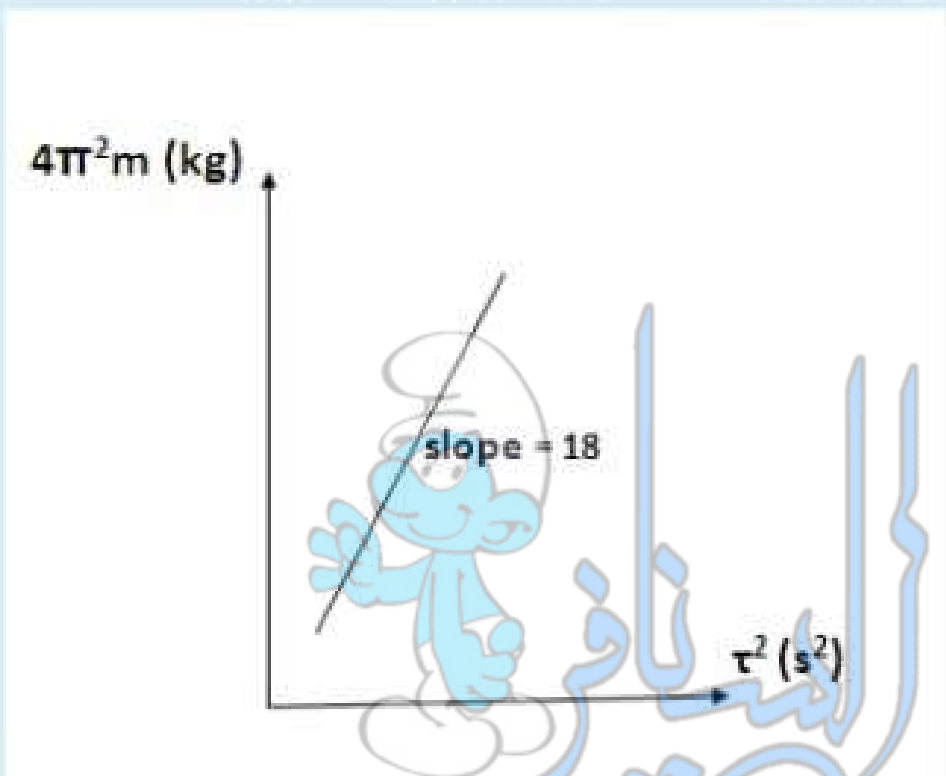
a. 8.6 N/m ☐

b. 6 N/m ☐

c. 12 N/m ☐

d. 10 N/m ☐

In Hook's law experiment, student draw the relation shown , the spring constant (k):



Select one:

- ☐ A. 10 N/m
- ☐ B. 3 N/m
- ☐ C. 18 N/m
- ☒ D. 9 N/m

The unit of spring constant is (kg/s^2) .

Select one:



True



False

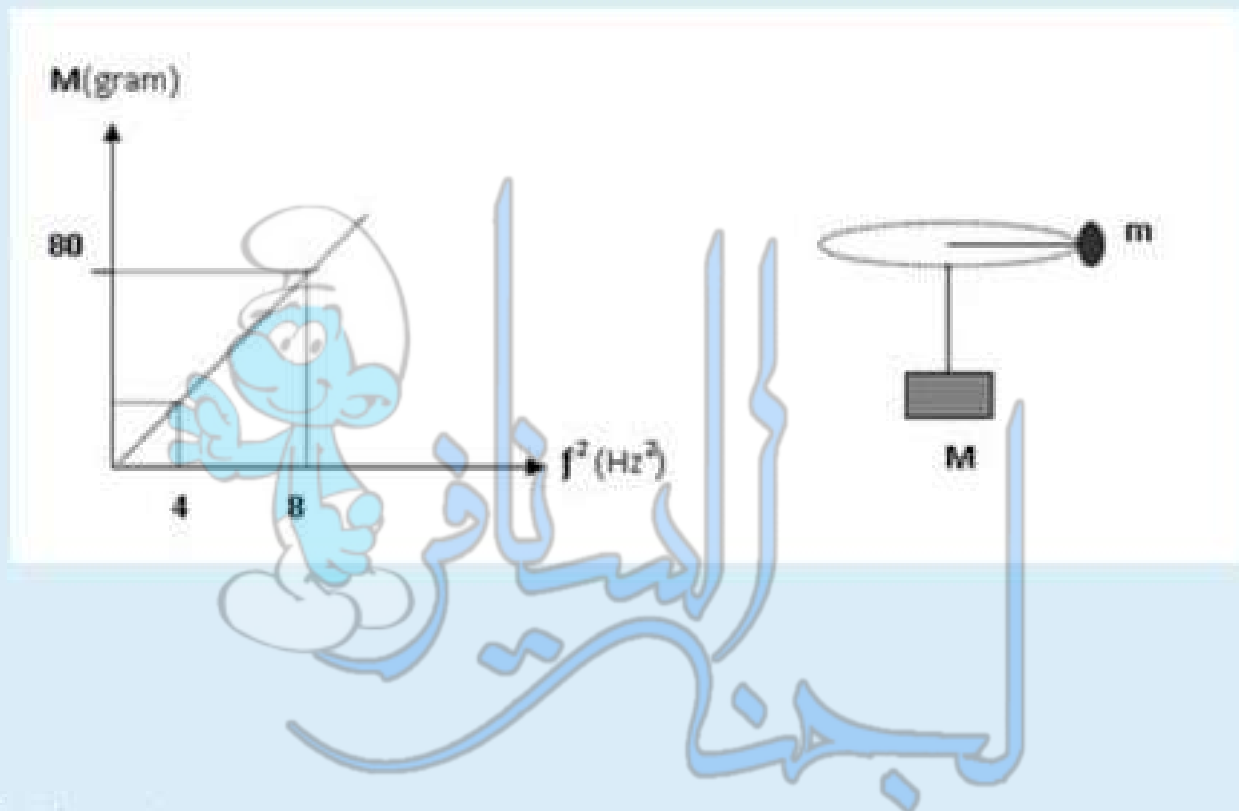


In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

Select one:

- ☐ a. The speed of the mass (M) will decrease
- ☐ b. The mass(M) will stop.
- ☐ c. The speed of the mass (M) will increase.
- ☐ d. The mass (M) will move in constant speed.

In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius ($R=0.3$ m). A plotted data are shown in the figure bellow. The mass (m) of the rotational body is:



Select one:

- ☐ a. Non of above
- ☒ b. 8.45 gram
- ☐ c. 4.22 gram
- ☐ d. 6.34 gram
- ☐ e. 12.7 gram

In a gravitational potential energy experiment, if the initial height of the mass (m) is ($h_i = 1 \text{ m}$) and its mass (20g).

The final kinetic energy ($(K.E)_f$), when the mass (m) reaches the height ($h_f = 80\text{cm}$) is:

Select one:

- ☐ a. **0.04 J**
- ☒ b. **2 J**
- ☐ c. **0.3 J**
- ☐ d. **0.4 J**



Square-shaped plate with side length (L).The uncertainty (error) in it area is:

Select one:

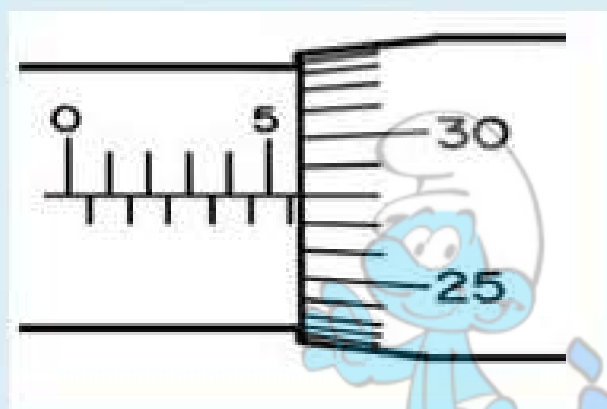
☐ a. $4L.\Delta L$

☐ b. $4L^2.\Delta L$

☐ c. $2L^2.\Delta L$

☒ d. $2L.\Delta L$

In an experiment to measure the density of a small metallic sphere, the diameter of the sphere (d) was measured by micrometer shown in the figure, and the mass of the sphere M was $M \pm \Delta M = (2.1 \pm 0.05) \text{ g}$. calculate the density of the sphere, density $\pm$ error density



Select one:

- ☒ a. 20792 ± 602.97
- ☐ b. 10762 ± 302.96
- ☐ c. 30552 ± 56.77
- ☐ d. Non of above
- ☐ e. 24893 ± 652.98

A cube with length (L), the error in the
:surface area



إختتر أحد الخيارات:

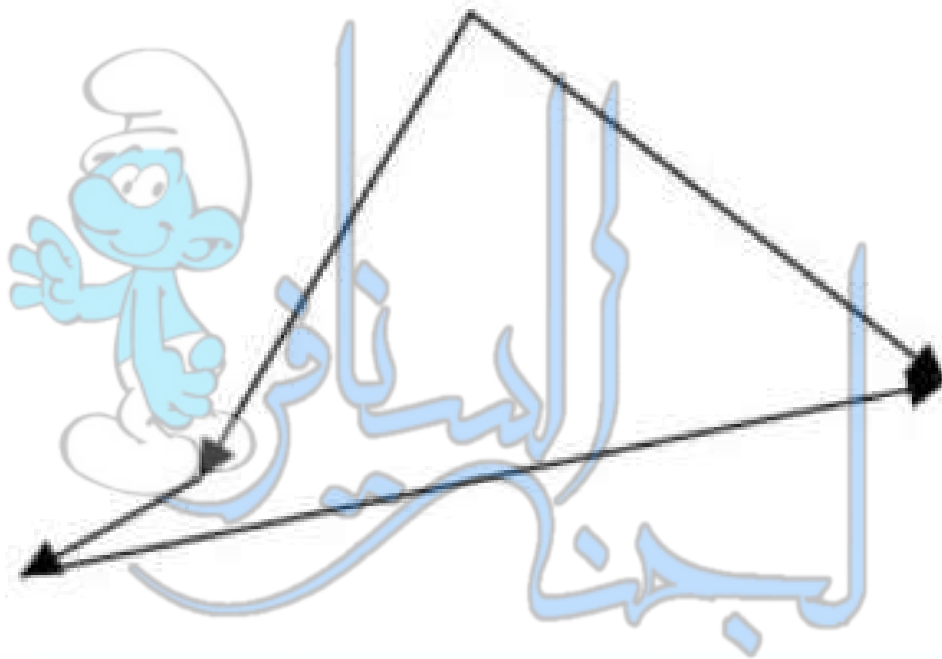
A. $12\Delta L.L$ ☐

B. $6\Delta L.L$ ☐

C. $12\Delta L.L^2$ ☐

D. $6\Delta L.L^2$ ☐

In the shown vector addition, the resultant of forces is zero



Select one:

☐ True

☒ False

In a balanced system on the force table, adding 30g to each strapless will remain balanced

Select one:

☒ True

☐ False



In a vector experiment, the ring being large or small does not affect the equilibrium

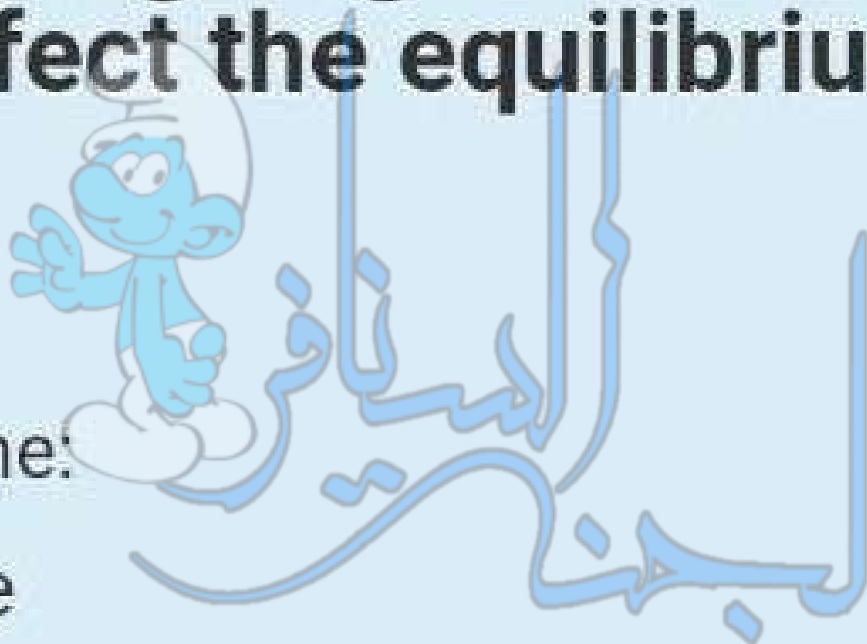
Select one:



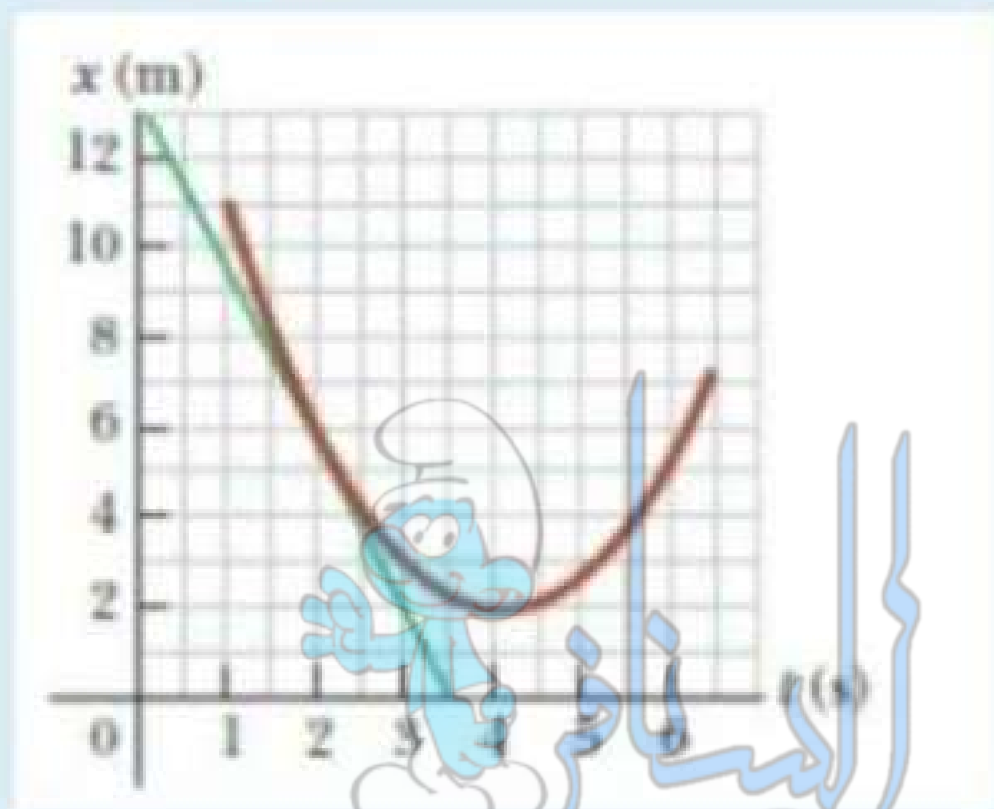
True



False



A position-time graph for a particle moving along the x-axis is shown in the next figure. What is the velocity at $t = 2$ s?



Select one:

- ☐ a. -3.8 m/s
- ☐ b. 3.71 m/s
- ☐ c. Cannot be determined from the graph
- ☐ d. 3.8 m/s
- ☐ e. -3.71 m/s

**A balanced system of three forces,
if the sum of (F2) and (F3) equals**

then the magnitude of (F1) ,(0.89N)

((0.89)N

إختتر واحداً:

صح ☐

خطأ ☒

A cube with length (L), the error in the
:surface area



إختر أحد الخيارات:

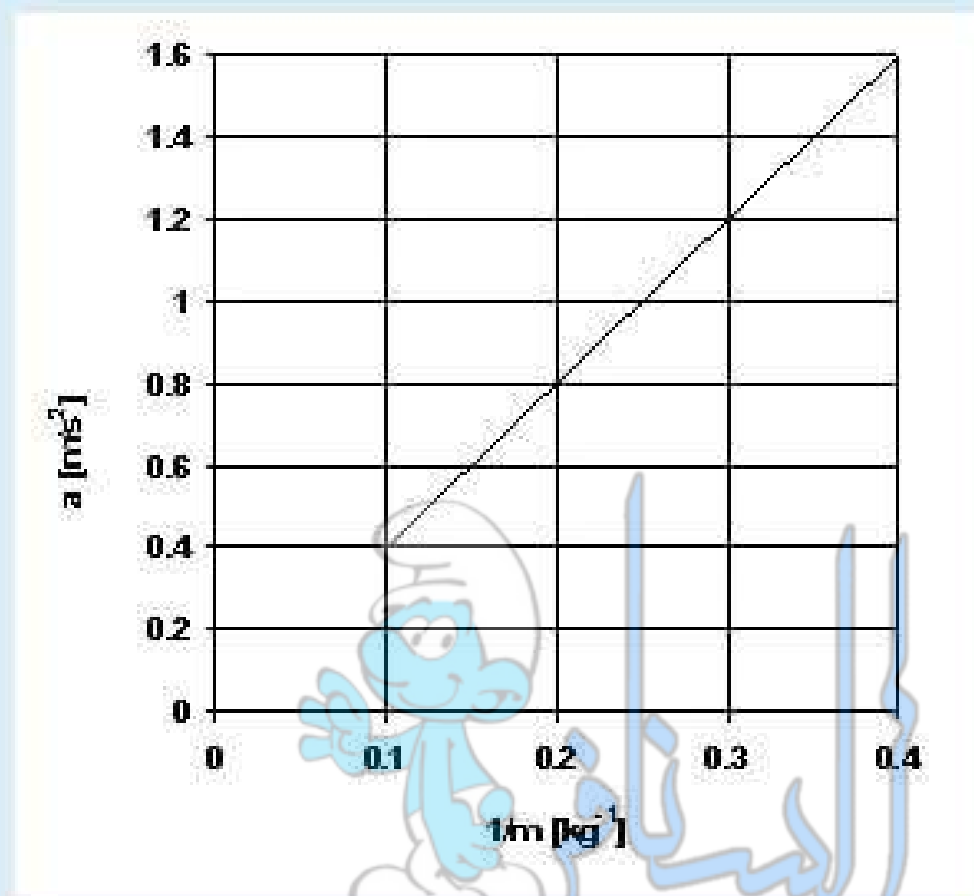
A. $12\Delta L.L$ ☐

B. $6\Delta L.L$ ☐

C. $12\Delta L.L^2$ ☐

D. $6\Delta L.L^2$ ☐

In the Newton's second law experiment, the relation between the acceleration (a) and the total mass (m) with a constant force (F) is shown in Figure. The applied force on the system (in N) is:



Select one:

- ☐ a. 12 N
- ☐ b. 8 N
- ☐ c. Non of above
- ☐ d. 16 N
- ☒ e. 4 N ✓

The correct answer is: 4 N

If F is the force, x the displacement, and k a spring constant, for simple harmonic motion, F is given by:

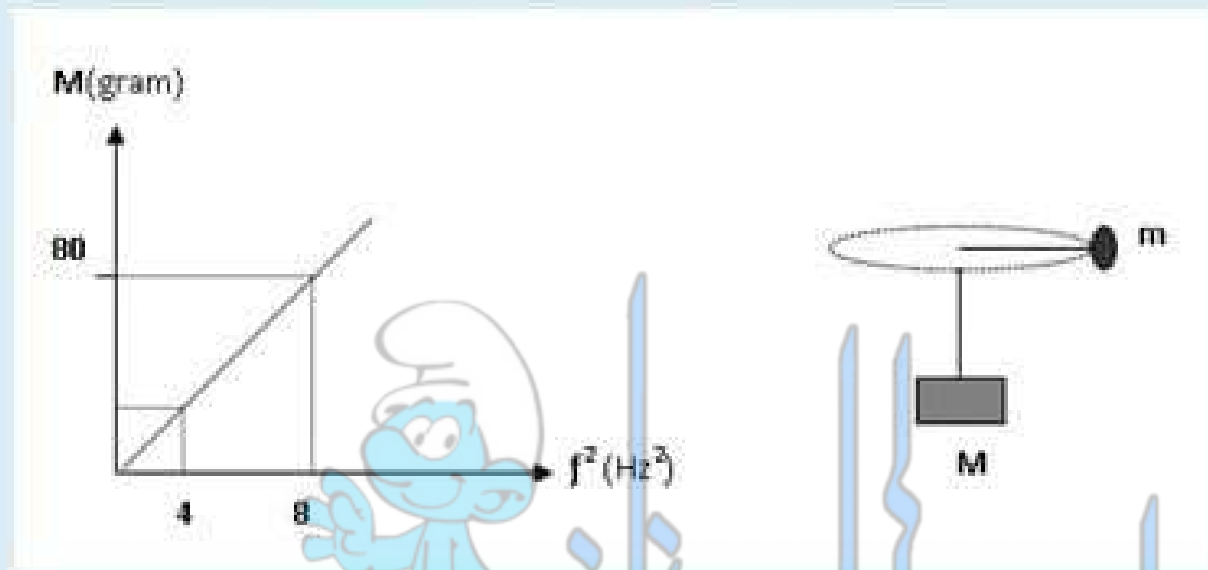
Select one:

- ☐ a. $F=kx$
- ☐ b. $F = -kx$
- ☒ c. $F=-k/x$ 
- ☐ d. $F = -kx^2$
- ☐ e. Non of above



The correct answer is: $F = -kx$

In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius ($R=0.3$ m). A plotted data are shown in the figure bellow. The mass (m) of the rotational body is:

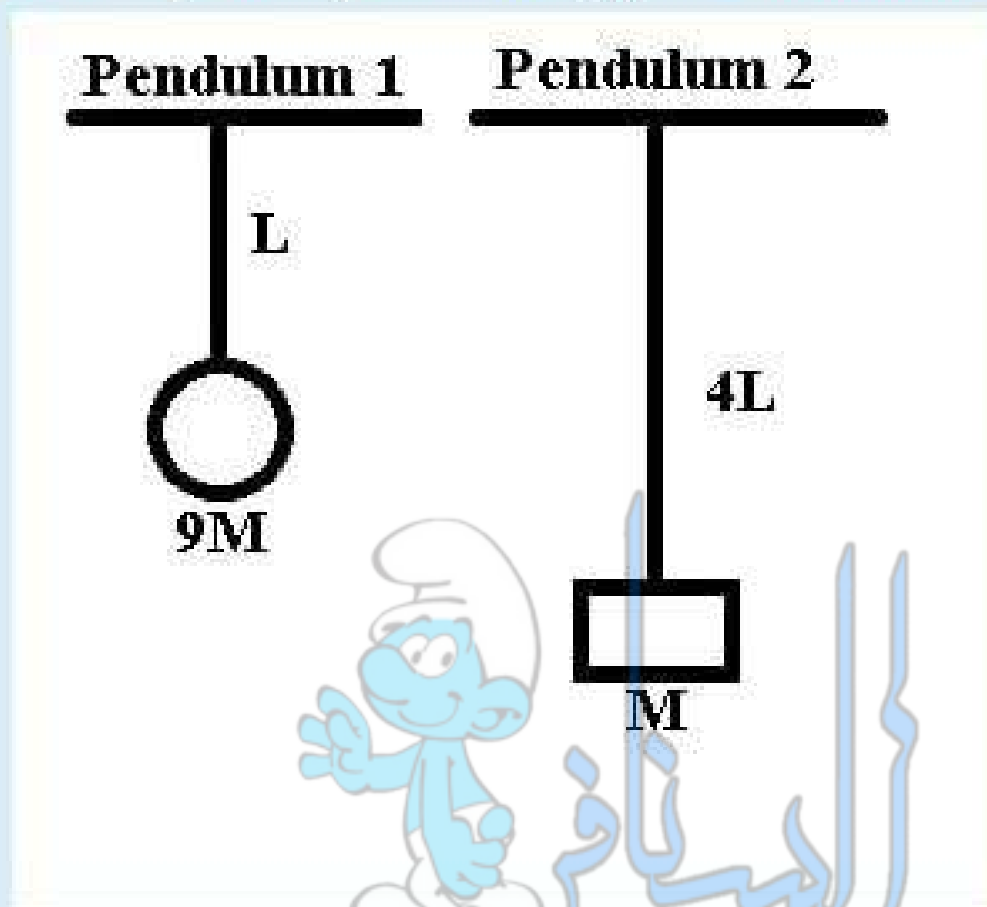


Select one:

- ☐ a. 6.34 gram
- ☒ b. 8.44 gram ✓
- ☐ c. 12.7 gram
- ☐ d. Non of above
- ☐ e. 4.22 gram

The correct answer is: 8.44 gram

In the simple pendulum experiment, the following figures show two pendulums. If the periodic time is T_1 for the Pendulum (1), and the periodic time is T_2 for the pendulum (2), then the relation between T_1 and T_2 is



Select one:

- ☐ a. $T_2 = 2.45T_1$
- ☐ b. Non of above
- ☐ c. $T_2 = 2.24T_1$
- ☒ d. $T_2 = 2T_1$ ✓
- ☐ e. $T_2 = 1.73T_1$

The correct answer is: $T_2 = 2T_1$

Micrometer reading is (8.74 mm), if we turned it (8) turns and (3/5) of turn, towards opening direction, its reading will be:

Select one:

- ☒ A. 5.34 mm
- ☐ B. 13.04 mm
- ☐ C. 4.54 mm
- ☐ D. 12.14 mm

A student used flexible ruler with maximum range (30 cm) to find circumference of circular table. If the radius of the table (80cm), then the uncertainty in circumference (ΔC) is

إختتر أحد الخيارات:

A. 17 mm ☒

B. 8.5 mm ☐

C. 10.5 mm ☐

D. 5.5 mm ☐

In the simple harmonic oscillator, a mass of (40g) is hanged in a spring, if the extension of the spring is (4 cm) and if the mass drawn to (3cm) from equilibrium point and oscillated 25 times, the total time of an oscillation of 20 times is (in sec).

Select one:

- ☐ a. 9.93
- ☐ b. 11.91
- ☐ c. Non of above
- ☐ d. 5.96
- ☒ e. 7.95 ✓



The correct answer is: 7.95

In a vector experiment, the ring being large or small does not affect the equilibrium

Select one:



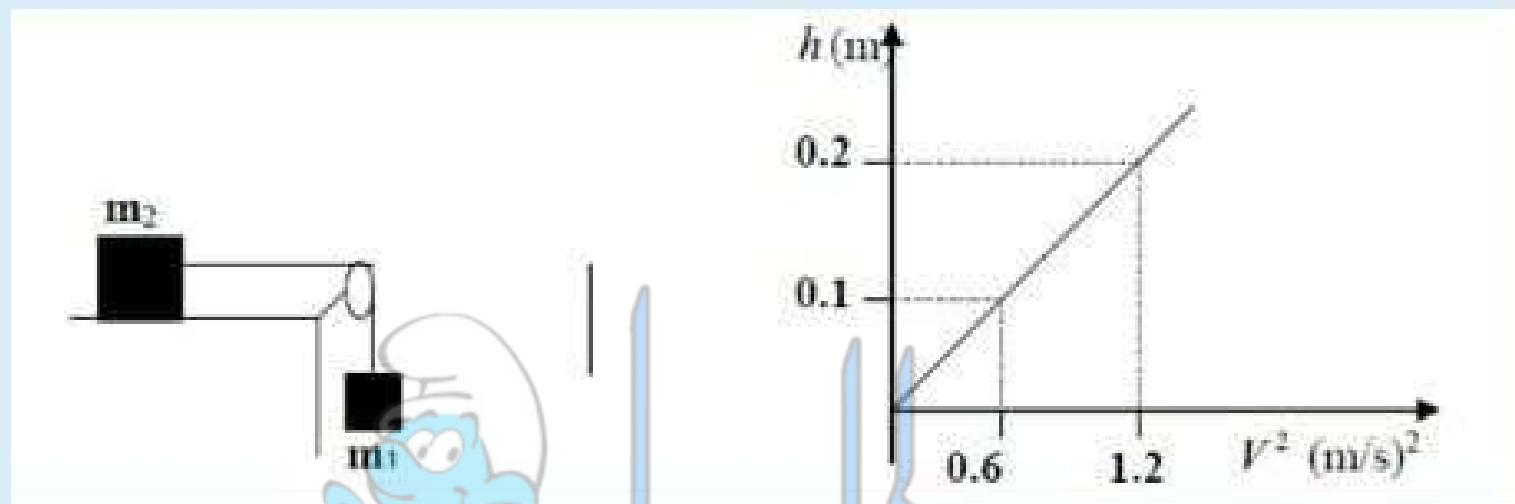
True



False



In the conservation of energy experiment, a student obtained data plotted in the following figure. If the masses $m_1 = 0.15 \text{ kg}$ and $m_2 = 0.2 \text{ kg}$, the acceleration of gravity (in m/s^2) is:

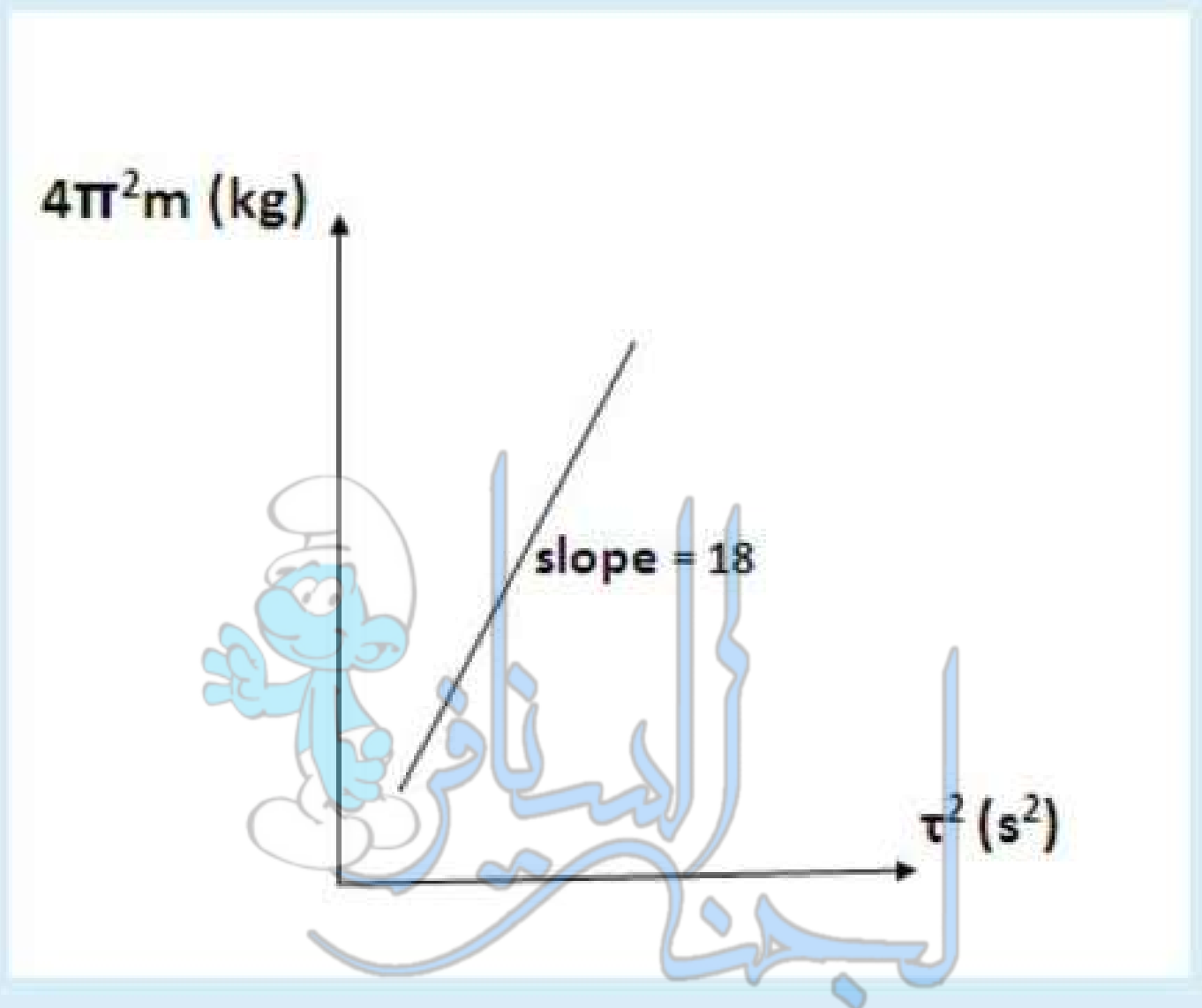


Select one:

- ☐ a. Non of above
- ☐ b. 11
- ☒ c. 7 ✓
- ☐ d. 9
- ☐ e. 13

The correct answer is: 7

In Hook's law experiment, student draw the relation shown , the spring constant (k):



Select one:

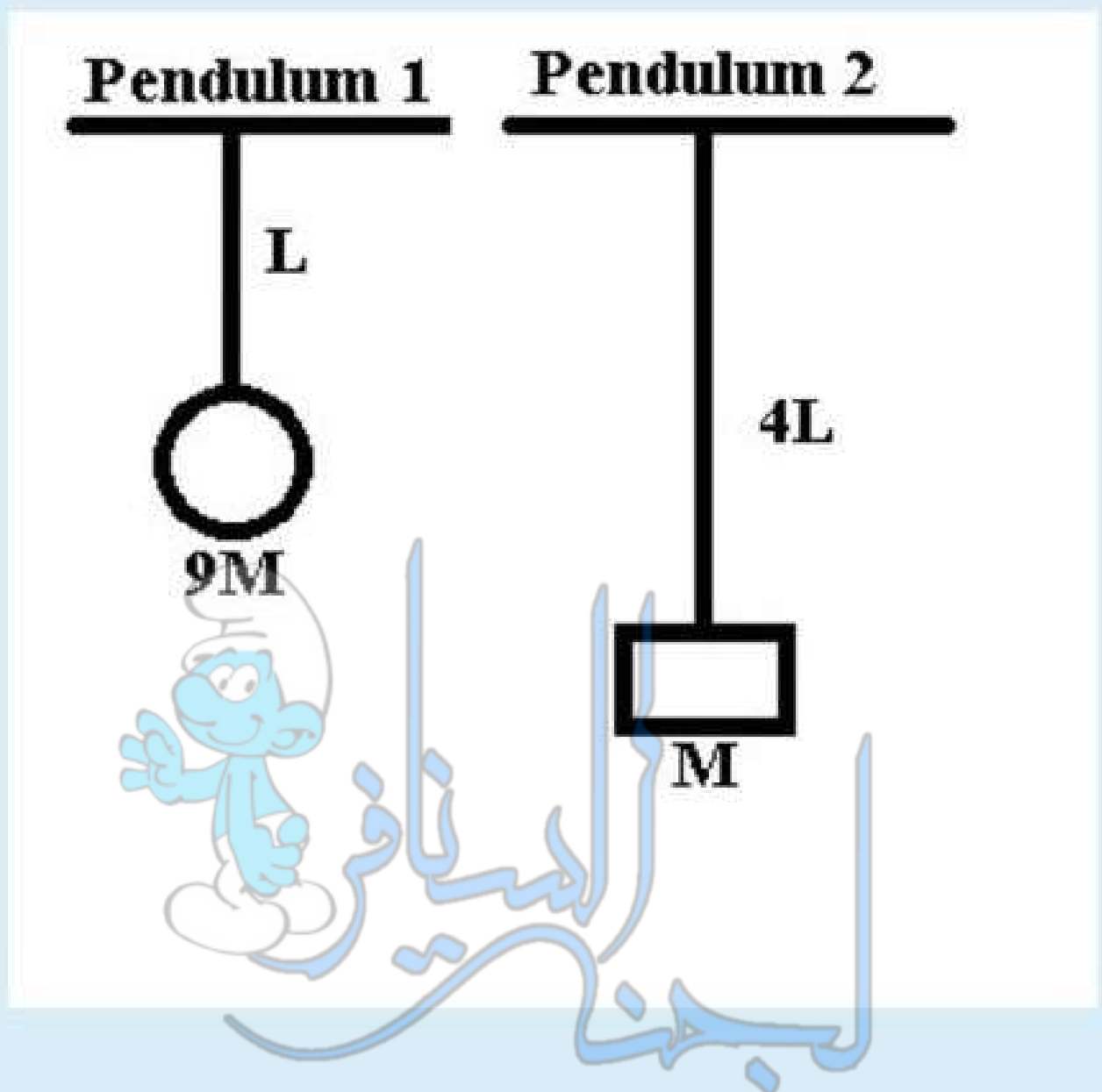
- ☐ A. 3 N/m
- ☒ B. 18 N/m
- ☐ C. 10 N/m
- ☐ D. 9 N/m

we can say about (μ_s) and (μ_k) that :

Select one:

- ☐ A. $-1 \geq (\mu_s) \text{ AND } (\mu_k) \geq 1$
- ☐ B. $(\mu_s) > (\mu_k)$
- ☐ C. $(\mu_s) < (\mu_k)$
- ☐ D. $(\mu_s) \text{ AND } (\mu_k) \geq 1$

The two pendulums shown have the same
:



Select one:

- ☐ a. acceleration of gravity
- ☒ b. predic time
- ☐ c. angular frecuanacy
- ☐ d. all of them
- ☐ e. Non of above